

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.2	6.7	1.6	0.4	0	77.1
U2 - Beta(0.5, 0.5)	14.8	6.9	1.4	0.1	0	76.8
U2 - Beta(0.7, 0.3)	14.6	6.8	1.5	0.1	0	77
U2 - Beta(1, 1)	15.3	6.7	1.1	0.1	0	76.8
U3 - Beta(0.3, 0.7)	15.8	5.5	1.2	0.2	0	77.3
U3 - Beta(0.5, 0.5)	16.1	5.6	1	0.2	0	77.1
U3 - Beta(0.7, 0.3)	16.7	5.3	0.9	0	0	77.1
U3 - Beta(1, 1)	17.2	5.4	0.8	0	0	76.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	3.3	1.7	0.6	0.2	-
U2 - Beta(0.5, 0.5)	11.3	3.3	1.7	0.6	0.2	-
U2 - Beta(0.7, 0.3)	11.4	3.3	1.6	0.6	0.2	-
U2 - Beta(1, 1)	11.3	3.3	1.7	0.6	0.2	-
U3 - Beta(0.3, 0.7)	11.3	3.3	1.7	0.6	0.2	-
U3 - Beta(0.5, 0.5)	11.3	3.4	1.7	0.6	0.2	-
U3 - Beta(0.7, 0.3)	11.3	3.4	1.7	0.6	0.2	-
U3 - Beta(1, 1)	11.4	3.3	1.7	0.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.3	1.7	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3.3	1.7	0.6	0.3	0.1	-
U3 - Beta(1, 1)	3.3	1.7	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						608.3
U2 - Beta(0.5, 0.5)						607.7
U2 - Beta(0.7, 0.3)						607.7
U2 - Beta(1, 1)						608.3
U3 - Beta(0.3, 0.7)						608.3
U3 - Beta(0.5, 0.5)						608.4
U3 - Beta(0.7, 0.3)						607.8
U3 - Beta(1, 1)						608.9
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34.4	16.5	31.7	10.2	2.4	4.8
U2 - Beta(0.5, 0.5)	33.9	17.9	31.5	9.4	2.3	5
U2 - Beta(0.7, 0.3)	33.8	16.7	32.4	10.1	2.1	4.9
U2 - Beta(1, 1)	34.5	16.9	31.9	9.7	2.2	4.8
U3 - Beta(0.3, 0.7)	36.6	17.3	30.6	9.1	1.5	4.9
U3 - Beta(0.5, 0.5)	37.2	18.4	30.2	8	1.4	4.8
U3 - Beta(0.7, 0.3)	37.8	18.2	29.1	8.4	1.6	4.9
U3 - Beta(1, 1)	38.5	18.2	29	7.9	1.5	4.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	5.2	7.2	3.7	1.6	-
U2 - Beta(0.5, 0.5)	11.5	5.4	7	3.6	1.6	-
U2 - Beta(0.7, 0.3)	11.6	5.3	7.1	3.6	1.5	-
U2 - Beta(1, 1)	11.6	5.2	7.1	3.7	1.5	-
U3 - Beta(0.3, 0.7)	11.5	5.2	7.2	3.7	1.5	-
U3 - Beta(0.5, 0.5)	11.5	5.3	7.1	3.6	1.6	-
U3 - Beta(0.7, 0.3)	11.6	5.3	7.2	3.5	1.6	-
U3 - Beta(1, 1)	11.6	5.2	7	3.7	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	2.5	2	1	0.4	-
U2 - Beta(0.5, 0.5)	3.3	2.5	1.9	1	0.4	-
U2 - Beta(0.7, 0.3)	3.4	2.5	2	0.9	0.4	-
U2 - Beta(1, 1)	3.4	2.5	2	1	0.4	-
U3 - Beta(0.3, 0.7)	3.3	2.5	2	1	0.4	-
U3 - Beta(0.5, 0.5)	3.3	2.5	2	1	0.4	-
U3 - Beta(0.7, 0.3)	3.4	2.5	2	0.9	0.4	-
U3 - Beta(1, 1)	3.4	2.5	1.9	1	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1099.8
U2 - Beta(0.5, 0.5)						1098.1
U2 - Beta(0.7, 0.3)						1098.4
U2 - Beta(1, 1)						1098.5
U3 - Beta(0.3, 0.7)						1098.9
U3 - Beta(0.5, 0.5)						1099
U3 - Beta(0.7, 0.3)						1098.6
U3 - Beta(1, 1)						1099

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.9	41.7	18.5	7.7	1.6	18.6
U2 - Beta(0.5, 0.5)	12.6	41.1	18.6	7.8	1.5	18.4
U2 - Beta(0.7, 0.3)	13	40.6	20	6	1.7	18.7
U2 - Beta(1, 1)	12.6	40.9	19.3	6.8	1.7	18.7
U3 - Beta(0.3, 0.7)	14.4	42.2	17.4	6.4	1.2	18.4
U3 - Beta(0.5, 0.5)	13	43.9	16.9	6.2	1.5	18.5
U3 - Beta(0.7, 0.3)	14.7	43.4	16.8	5.3	1.3	18.5
U3 - Beta(1, 1)	14.7	44.7	15.2	5.5	1.4	18.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8.6	7.5	6	3.5	1.5	-
U2 - Beta(0.5, 0.5)	8.6	7.7	5.9	3.4	1.5	-
U2 - Beta(0.7, 0.3)	8.6	7.7	6	3.3	1.5	-
U2 - Beta(1, 1)	8.6	7.7	6	3.3	1.5	-
U3 - Beta(0.3, 0.7)	8.5	7.5	6.1	3.4	1.6	-
U3 - Beta(0.5, 0.5)	8.6	7.7	6	3.3	1.5	-
U3 - Beta(0.7, 0.3)	8.5	7.7	6	3.3	1.5	-
U3 - Beta(1, 1)	8.6	7.8	5.9	3.3	1.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	3.3	1.7	1.1	0.4	-
U2 - Beta(0.5, 0.5)	2.3	3.3	1.6	1.1	0.4	-
U2 - Beta(0.7, 0.3)	2.3	3.3	1.7	1.1	0.4	-
U2 - Beta(1, 1)	2.3	3.3	1.7	1	0.4	-
U3 - Beta(0.3, 0.7)	2.3	3.2	1.7	1.1	0.5	-
U3 - Beta(0.5, 0.5)	2.3	3.3	1.7	1	0.5	-
U3 - Beta(0.7, 0.3)	2.3	3.3	1.7	1	0.4	-
U3 - Beta(1, 1)	2.3	3.3	1.6	1.1	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1005.6
U2 - Beta(0.5, 0.5)						1005.3
U2 - Beta(0.7, 0.3)						1005.4
U2 - Beta(1, 1)						1005.4
U3 - Beta(0.3, 0.7)						1007
U3 - Beta(0.5, 0.5)						1005.1
U3 - Beta(0.7, 0.3)						1005
U3 - Beta(1, 1)						1005.5
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.5	8.5	22.8	43.1	15.4	3.7
U2 - Beta(0.5, 0.5)	7.2	8.5	22.8	43.6	14.3	3.6
U2 - Beta(0.7, 0.3)	7.6	8.9	21.4	43.8	14.5	3.8
U2 - Beta(1, 1)	8.6	9.1	21.2	43	14.4	3.7
U3 - Beta(0.3, 0.7)	9.3	9.6	22.8	41.3	13.5	3.5
U3 - Beta(0.5, 0.5)	11.3	9.1	21.8	42	12.3	3.5
U3 - Beta(0.7, 0.3)	13	9.9	19.4	41.1	13	3.6
U3 - Beta(1, 1)	13.6	9.6	19.8	40.5	12.9	3.6
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	4.2	6.7	8.1	5.9	-
U2 - Beta(0.5, 0.5)	4.4	4.3	6.7	8.2	5.8	-
U2 - Beta(0.7, 0.3)	4.5	4.3	6.7	8.2	5.7	-
U2 - Beta(1, 1)	4.4	4.3	6.7	8.1	5.8	-
U3 - Beta(0.3, 0.7)	4.3	4.2	6.7	8.2	5.8	-
U3 - Beta(0.5, 0.5)	4.4	4.4	6.7	8.1	5.8	-
U3 - Beta(0.7, 0.3)	4.4	4.4	6.7	8.3	5.6	-
U3 - Beta(1, 1)	4.4	4.3	6.7	8.1	5.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	2.8	2.1	2.7	1.4	-
U2 - Beta(0.5, 0.5)	0.6	2.8	2.1	2.7	1.4	-
U2 - Beta(0.7, 0.3)	0.7	2.8	2.1	2.7	1.3	-
U2 - Beta(1, 1)	0.6	2.8	2.1	2.6	1.4	-
U3 - Beta(0.3, 0.7)	0.6	2.8	2.1	2.7	1.4	-
U3 - Beta(0.5, 0.5)	0.6	2.8	2.1	2.6	1.4	-
U3 - Beta(0.7, 0.3)	0.7	2.8	2.1	2.7	1.3	-
U3 - Beta(1, 1)	0.7	2.8	2.1	2.6	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1089.3
U2 - Beta(0.5, 0.5)						1089.3
U2 - Beta(0.7, 0.3)						1088.9
U2 - Beta(1, 1)						1089.8
U3 - Beta(0.3, 0.7)						1090
U3 - Beta(0.5, 0.5)						1090.2
U3 - Beta(0.7, 0.3)						1090.4
U3 - Beta(1, 1)						1088.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.8	34.2	24.3	19.8	11.8	6.1
U2 - Beta(0.5, 0.5)	3.2	35.4	25.4	20	9.9	6.1
U2 - Beta(0.7, 0.3)	4	36.4	25.6	19.4	8.5	6.1
U2 - Beta(1, 1)	4.7	37.3	25.3	18.4	8.2	6.1
U3 - Beta(0.3, 0.7)	5.9	39.1	24.3	17.3	7.3	6.1
U3 - Beta(0.5, 0.5)	6.2	41.2	25	15.9	5.6	6.1
U3 - Beta(0.7, 0.3)	7.3	42.4	23.7	15.7	4.8	6.1
U3 - Beta(1, 1)	9.1	42.7	23.4	14.4	4.3	6.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	6.4	6.8	5.8	5.3	-
U2 - Beta(0.5, 0.5)	4.6	6.6	6.7	5.8	5.2	-
U2 - Beta(0.7, 0.3)	4.6	6.7	6.7	5.8	5.1	-
U2 - Beta(1, 1)	4.6	6.7	6.7	5.8	5.1	-
U3 - Beta(0.3, 0.7)	4.5	6.2	6.8	5.8	5.4	-
U3 - Beta(0.5, 0.5)	4.5	6.5	6.8	5.8	5.2	-
U3 - Beta(0.7, 0.3)	4.5	6.6	6.7	5.9	5.1	-
U3 - Beta(1, 1)	4.6	6.6	6.8	5.8	5.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.5	2.1	2.2	1.8	-
U2 - Beta(0.5, 0.5)	0.7	3.5	2.1	2.2	1.8	-
U2 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.8	-
U2 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.8	-
U3 - Beta(0.3, 0.7)	0.7	3.4	2.2	2.2	1.9	-
U3 - Beta(0.5, 0.5)	0.7	3.5	2.2	2.2	1.8	-
U3 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.8	-
U3 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1015.6
U2 - Beta(0.5, 0.5)						1014.5
U2 - Beta(0.7, 0.3)						1014.3
U2 - Beta(1, 1)						1014
U3 - Beta(0.3, 0.7)						1016.3
U3 - Beta(0.5, 0.5)						1015.3
U3 - Beta(0.7, 0.3)						1014
U3 - Beta(1, 1)						1014.7
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	66.4	11.6	2.6	1.3	0.1	18
U2 - Beta(0.5, 0.5)	66.7	11.5	2.8	0.8	0.3	17.9
U2 - Beta(0.7, 0.3)	65.8	12.4	2.8	0.8	0.2	18
U2 - Beta(1, 1)	66.3	12.3	2.4	1	0.2	17.8
U3 - Beta(0.3, 0.7)	67.9	11.4	2.1	0.6	0.2	17.8
U3 - Beta(0.5, 0.5)	67.2	11.7	2.6	0.5	0.1	17.9
U3 - Beta(0.7, 0.3)	67.3	11.8	2.3	0.3	0.1	18.2
U3 - Beta(1, 1)	68.7	10.7	2.2	0.4	0.2	17.8
Mean administrations						
U2 - Beta(0.3, 0.7)	19.7	3.8	2.1	0.9	0.3	-
U2 - Beta(0.5, 0.5)	19.7	3.8	2.2	0.8	0.3	-
U2 - Beta(0.7, 0.3)	19.7	3.8	2.2	0.9	0.2	-
U2 - Beta(1, 1)	19.7	3.8	2.1	0.9	0.2	-
U3 - Beta(0.3, 0.7)	19.7	3.8	2.1	0.9	0.3	-
U3 - Beta(0.5, 0.5)	19.7	3.9	2.1	0.9	0.3	-
U3 - Beta(0.7, 0.3)	19.7	3.8	2.1	0.8	0.3	-
U3 - Beta(1, 1)	19.7	3.8	2.2	0.9	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.5	1.5	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	5.4	1.5	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	5.4	1.5	0.6	0.3	0.1	-
U2 - Beta(1, 1)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	5.5	1.5	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	5.4	1.5	0.6	0.3	0.1	-
U3 - Beta(1, 1)	5.4	1.5	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1036.7
U2 - Beta(0.5, 0.5)						1037.8
U2 - Beta(0.7, 0.3)						1037.9
U2 - Beta(1, 1)						1037.9
U3 - Beta(0.3, 0.7)						1037.6
U3 - Beta(0.5, 0.5)						1038.3
U3 - Beta(0.7, 0.3)						1035.3
U3 - Beta(1, 1)						1037.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	24.8	36.3	20.6	10.4	2.4	5.5
U2 - Beta(0.5, 0.5)	25.1	38	19.6	9.6	2.1	5.6
U2 - Beta(0.7, 0.3)	25.3	37.9	18.9	10.2	2.2	5.5
U2 - Beta(1, 1)	24.8	37.3	20.7	9.5	2.2	5.5
U3 - Beta(0.3, 0.7)	26	38.7	19.1	8.9	1.8	5.5
U3 - Beta(0.5, 0.5)	26.8	41.1	16.6	8.4	1.6	5.5
U3 - Beta(0.7, 0.3)	27	41.2	16.5	8.2	1.6	5.5
U3 - Beta(1, 1)	27.5	40.9	16.8	7.7	1.6	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	10.9	6.9	5.6	3.6	2	-
U2 - Beta(0.5, 0.5)	10.9	7	5.5	3.6	2	-
U2 - Beta(0.7, 0.3)	10.9	7	5.6	3.6	1.9	-
U2 - Beta(1, 1)	10.9	6.9	5.6	3.7	2	-
U3 - Beta(0.3, 0.7)	10.8	6.9	5.6	3.7	2	-
U3 - Beta(0.5, 0.5)	10.9	7.1	5.5	3.6	1.9	-
U3 - Beta(0.7, 0.3)	10.9	7	5.6	3.6	1.9	-
U3 - Beta(1, 1)	10.9	7	5.5	3.6	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.8	1.4	1	0.5	-
U2 - Beta(0.5, 0.5)	3.1	2.8	1.3	1	0.5	-
U2 - Beta(0.7, 0.3)	3.1	2.8	1.4	1	0.5	-
U2 - Beta(1, 1)	3	2.8	1.4	1	0.5	-
U3 - Beta(0.3, 0.7)	3	2.8	1.4	1	0.5	-
U3 - Beta(0.5, 0.5)	3	2.9	1.4	1	0.5	-
U3 - Beta(0.7, 0.3)	3.1	2.8	1.4	1	0.5	-
U3 - Beta(1, 1)	3.1	2.8	1.3	1.1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1114.1
U2 - Beta(0.5, 0.5)						1113.9
U2 - Beta(0.7, 0.3)						1114
U2 - Beta(1, 1)						1114.6
U3 - Beta(0.3, 0.7)						1114.2
U3 - Beta(0.5, 0.5)						1114.3
U3 - Beta(0.7, 0.3)						1114.1
U3 - Beta(1, 1)						1114.6
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.7	6.5	1.6	0.7	0.2	63.3
U2 - Beta(0.5, 0.5)	27.8	6.5	1.8	0.9	0.1	62.9
U2 - Beta(0.7, 0.3)	28.2	6.5	1.4	0.5	0.2	63.2
U2 - Beta(1, 1)	29.5	5.7	1.3	0.7	0.1	62.7
U3 - Beta(0.3, 0.7)	29.3	5.4	1.5	0.5	0.2	63.1
U3 - Beta(0.5, 0.5)	29.4	5.1	1.5	0.7	0.2	63.1
U3 - Beta(0.7, 0.3)	29.8	6.1	1	0.6	0.2	62.3
U3 - Beta(1, 1)	29.6	5.7	1.1	0.6	0.2	62.8
Mean administrations						
U2 - Beta(0.3, 0.7)	14	3.3	1.5	0.5	0.1	-
U2 - Beta(0.5, 0.5)	14	3.3	1.5	0.5	0.1	-
U2 - Beta(0.7, 0.3)	14	3.2	1.5	0.5	0.1	-
U2 - Beta(1, 1)	14.1	3.3	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	14	3.3	1.5	0.5	0.1	-
U3 - Beta(0.5, 0.5)	14	3.3	1.5	0.5	0.1	-
U3 - Beta(0.7, 0.3)	14	3.2	1.5	0.5	0.1	-
U3 - Beta(1, 1)	14	3.3	1.5	0.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	1.5	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	4.1	1.6	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	4.2	1.5	0.5	0.2	0	-
U2 - Beta(1, 1)	4.2	1.5	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	4.2	1.6	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	4.2	1.5	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	4.2	1.5	0.5	0.2	0	-
U3 - Beta(1, 1)	4.2	1.5	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						705.2
U2 - Beta(0.5, 0.5)						704.7
U2 - Beta(0.7, 0.3)						704.5
U2 - Beta(1, 1)						706
U3 - Beta(0.3, 0.7)						705.6
U3 - Beta(0.5, 0.5)						705.8
U3 - Beta(0.7, 0.3)						704.7
U3 - Beta(1, 1)						704.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	9.8	17.8	12.3	22.5	29.7
U2 - Beta(0.5, 0.5)	8.7	10.9	17.2	11.9	21.9	29.4
U2 - Beta(0.7, 0.3)	11.4	10.3	15.9	11	21.9	29.5
U2 - Beta(1, 1)	10.6	12.7	14.7	10.9	21.6	29.5
U3 - Beta(0.3, 0.7)	12.4	10.8	15.9	10.2	21.2	29.5
U3 - Beta(0.5, 0.5)	14.3	10.2	15.3	10	20.6	29.6
U3 - Beta(0.7, 0.3)	15.7	11.9	12.9	11.3	18.7	29.5
U3 - Beta(1, 1)	18.5	10.7	11.9	10.3	19.1	29.5
Mean administrations						
U2 - Beta(0.3, 0.7)	7.2	4.1	5.5	3.7	4.6	-
U2 - Beta(0.5, 0.5)	7.2	4.3	5.4	3.7	4.5	-
U2 - Beta(0.7, 0.3)	7.3	4.4	5.3	3.6	4.4	-
U2 - Beta(1, 1)	7.3	4.4	5.2	3.6	4.5	-
U3 - Beta(0.3, 0.7)	7.1	4.2	5.5	3.6	4.6	-
U3 - Beta(0.5, 0.5)	7.3	4.2	5.4	3.7	4.5	-
U3 - Beta(0.7, 0.3)	7.3	4.3	5.3	3.7	4.4	-
U3 - Beta(1, 1)	7.3	4.4	5.2	3.6	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	2.4	1.9	1.5	1.3	-
U2 - Beta(0.5, 0.5)	1.9	2.5	1.9	1.5	1.3	-
U2 - Beta(0.7, 0.3)	1.9	2.5	1.9	1.5	1.3	-
U2 - Beta(1, 1)	1.9	2.5	1.8	1.5	1.3	-
U3 - Beta(0.3, 0.7)	1.8	2.5	1.9	1.5	1.3	-
U3 - Beta(0.5, 0.5)	1.9	2.4	1.9	1.5	1.3	-
U3 - Beta(0.7, 0.3)	1.9	2.5	1.8	1.5	1.3	-
U3 - Beta(1, 1)	1.9	2.5	1.8	1.5	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						882.5
U2 - Beta(0.5, 0.5)						882.2
U2 - Beta(0.7, 0.3)						880.2
U2 - Beta(1, 1)						882.1
U3 - Beta(0.3, 0.7)						883.7
U3 - Beta(0.5, 0.5)						881.1
U3 - Beta(0.7, 0.3)						879.5
U3 - Beta(1, 1)						879.7

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	25	11	3.9	1.2	51
U2 - Beta(0.5, 0.5)	8	25.5	10.7	3.5	1.1	51.2
U2 - Beta(0.7, 0.3)	7.8	26.2	10.4	3.1	1.2	51.3
U2 - Beta(1, 1)	8	25.3	10.3	4	1.3	51.1
U3 - Beta(0.3, 0.7)	8.1	26.4	9.3	3.7	1.1	51.4
U3 - Beta(0.5, 0.5)	8.5	27.7	9	2.9	0.9	51
U3 - Beta(0.7, 0.3)	8.4	27.7	9.1	2.8	0.8	51.2
U3 - Beta(1, 1)	8.9	28.3	8	2.7	1	51.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	5.4	3.8	1.8	0.7	-
U2 - Beta(0.5, 0.5)	9.2	5.5	3.7	1.8	0.7	-
U2 - Beta(0.7, 0.3)	9.2	5.5	3.6	1.8	0.7	-
U2 - Beta(1, 1)	9.2	5.5	3.6	1.8	0.8	-
U3 - Beta(0.3, 0.7)	9.2	5.4	3.7	1.9	0.8	-
U3 - Beta(0.5, 0.5)	9.2	5.5	3.7	1.8	0.8	-
U3 - Beta(0.7, 0.3)	9.2	5.5	3.7	1.8	0.7	-
U3 - Beta(1, 1)	9.2	5.5	3.6	1.8	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	2.4	1	0.6	0.2	-
U2 - Beta(0.5, 0.5)	2.5	2.5	1	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.5	2.5	1	0.5	0.2	-
U2 - Beta(1, 1)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(0.3, 0.7)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(0.5, 0.5)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.5	2.4	1	0.6	0.2	-
U3 - Beta(1, 1)	2.5	2.5	1	0.6	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						775.5
U2 - Beta(0.5, 0.5)						776.6
U2 - Beta(0.7, 0.3)						776.2
U2 - Beta(1, 1)						775.8
U3 - Beta(0.3, 0.7)						776.4
U3 - Beta(0.5, 0.5)						776
U3 - Beta(0.7, 0.3)						775.4
U3 - Beta(1, 1)						775.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	64.2	18.3	2.8	0.1	0	14.6
U2 - Beta(0.5, 0.5)	63.9	18.9	2.7	0.1	0	14.4
U2 - Beta(0.7, 0.3)	63.5	19.1	2.9	0.1	0	14.4
U2 - Beta(1, 1)	63.1	19.7	2.5	0.1	0	14.6
U3 - Beta(0.3, 0.7)	65.4	17.8	2.1	0	0	14.7
U3 - Beta(0.5, 0.5)	64.9	18.2	2.2	0.1	0	14.6
U3 - Beta(0.7, 0.3)	66.6	16.7	2.3	0.1	0	14.3
U3 - Beta(1, 1)	67.8	16.4	1.2	0.1	0	14.5
Mean administrations						
U2 - Beta(0.3, 0.7)	18.1	5.7	2.8	0.7	0.1	-
U2 - Beta(0.5, 0.5)	18.1	5.7	2.8	0.7	0.1	-
U2 - Beta(0.7, 0.3)	18.1	5.7	2.9	0.7	0.1	-
U2 - Beta(1, 1)	18.1	5.7	2.8	0.7	0.1	-
U3 - Beta(0.3, 0.7)	18	5.7	2.9	0.7	0.1	-
U3 - Beta(0.5, 0.5)	18.1	5.7	2.8	0.7	0.1	-
U3 - Beta(0.7, 0.3)	18.1	5.7	2.9	0.7	0.1	-
U3 - Beta(1, 1)	18.2	5.7	2.8	0.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(0.5, 0.5)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(0.7, 0.3)	5.6	2.5	0.9	0.3	0	-
U2 - Beta(1, 1)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.3, 0.7)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.5, 0.5)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(0.7, 0.3)	5.6	2.5	0.9	0.3	0	-
U3 - Beta(1, 1)	5.6	2.5	0.9	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						997.8
U2 - Beta(0.5, 0.5)						998
U2 - Beta(0.7, 0.3)						997.4
U2 - Beta(1, 1)						997.7
U3 - Beta(0.3, 0.7)						997.1
U3 - Beta(0.5, 0.5)						998.1
U3 - Beta(0.7, 0.3)						998.2
U3 - Beta(1, 1)						998.4
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.6	13.4	12.4	37.5	17.6	5.5
U2 - Beta(0.5, 0.5)	14.1	14.8	11.5	36.5	17.4	5.7
U2 - Beta(0.7, 0.3)	14.1	14.7	11	37.7	16.9	5.6
U2 - Beta(1, 1)	16.7	13.4	10.2	37.1	17	5.6
U3 - Beta(0.3, 0.7)	18.5	14.1	11.2	34.6	16.1	5.5
U3 - Beta(0.5, 0.5)	20.6	14.2	9.3	35.5	15.1	5.3
U3 - Beta(0.7, 0.3)	21.8	13.8	8.5	34.4	15.8	5.7
U3 - Beta(1, 1)	24.4	14	8	32.6	15.8	5.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.8	4.7	5.6	7.3	6.6	-
U2 - Beta(0.5, 0.5)	4.9	4.8	5.6	7.2	6.5	-
U2 - Beta(0.7, 0.3)	4.9	4.9	5.5	7.4	6.4	-
U2 - Beta(1, 1)	5	4.9	5.5	7.2	6.5	-
U3 - Beta(0.3, 0.7)	4.8	4.7	5.6	7.4	6.5	-
U3 - Beta(0.5, 0.5)	4.9	4.8	5.6	7.3	6.5	-
U3 - Beta(0.7, 0.3)	4.9	5	5.5	7.3	6.4	-
U3 - Beta(1, 1)	4.9	5	5.4	7.2	6.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	2.9	1.8	2.8	1.5	-
U2 - Beta(0.5, 0.5)	0.9	3	1.8	2.7	1.5	-
U2 - Beta(0.7, 0.3)	0.9	3	1.8	2.7	1.5	-
U2 - Beta(1, 1)	0.9	3	1.7	2.7	1.5	-
U3 - Beta(0.3, 0.7)	0.9	2.9	1.8	2.8	1.5	-
U3 - Beta(0.5, 0.5)	0.9	3	1.8	2.8	1.4	-
U3 - Beta(0.7, 0.3)	0.9	3	1.8	2.7	1.5	-
U3 - Beta(1, 1)	0.9	3	1.7	2.7	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1062.3
U2 - Beta(0.5, 0.5)						1059.5
U2 - Beta(0.7, 0.3)						1059.5
U2 - Beta(1, 1)						1058.8
U3 - Beta(0.3, 0.7)						1062.6
U3 - Beta(0.5, 0.5)						1060.6
U3 - Beta(0.7, 0.3)						1057.3
U3 - Beta(1, 1)						1059.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.7	27.8	11.8	3.8	1.5	46.4
U2 - Beta(0.5, 0.5)	9.6	28	11.2	3.6	1.2	46.4
U2 - Beta(0.7, 0.3)	9.6	27.6	11.3	3.8	1.2	46.5
U2 - Beta(1, 1)	10	27.6	11.6	3.5	1	46.3
U3 - Beta(0.3, 0.7)	10.2	29.1	9.9	3.5	0.9	46.4
U3 - Beta(0.5, 0.5)	10.4	29	9.6	3.8	1.1	46.1
U3 - Beta(0.7, 0.3)	11.2	28.6	9.5	3.3	1.1	46.3
U3 - Beta(1, 1)	12.3	28.8	8.8	3.1	1	46
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	5.9	4.1	2	0.8	-
U2 - Beta(0.5, 0.5)	9	6	3.9	2	0.8	-
U2 - Beta(0.7, 0.3)	9	6	3.9	2	0.8	-
U2 - Beta(1, 1)	9	6	3.9	2	0.8	-
U3 - Beta(0.3, 0.7)	8.9	5.9	4	2	0.8	-
U3 - Beta(0.5, 0.5)	9	6	4	2	0.8	-
U3 - Beta(0.7, 0.3)	9	6.1	3.9	1.9	0.8	-
U3 - Beta(1, 1)	9	6	3.9	2	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	2.7	1.2	0.7	0.2	-
U2 - Beta(0.5, 0.5)	2.5	2.8	1.2	0.7	0.2	-
U2 - Beta(0.7, 0.3)	2.4	2.8	1.2	0.7	0.2	-
U2 - Beta(1, 1)	2.5	2.8	1.2	0.7	0.2	-
U3 - Beta(0.3, 0.7)	2.4	2.7	1.2	0.7	0.2	-
U3 - Beta(0.5, 0.5)	2.5	2.8	1.2	0.7	0.2	-
U3 - Beta(0.7, 0.3)	2.4	2.8	1.2	0.7	0.2	-
U3 - Beta(1, 1)	2.5	2.8	1.2	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						788.3
U2 - Beta(0.5, 0.5)						789.8
U2 - Beta(0.7, 0.3)						789.3
U2 - Beta(1, 1)						788.7
U3 - Beta(0.3, 0.7)						790
U3 - Beta(0.5, 0.5)						788
U3 - Beta(0.7, 0.3)						789.6
U3 - Beta(1, 1)						788.7
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14	15.3	16.4	26.3	13.3	14.7
U2 - Beta(0.5, 0.5)	18	14.8	14.3	26.1	12.6	14.2
U2 - Beta(0.7, 0.3)	18.5	14.4	14.8	25.7	12.2	14.4
U2 - Beta(1, 1)	20.3	14.6	13.2	26	11.1	14.8
U3 - Beta(0.3, 0.7)	21.2	16.4	12.4	24.8	11	14.2
U3 - Beta(0.5, 0.5)	25.4	15.1	11.3	24.5	9.4	14.3
U3 - Beta(0.7, 0.3)	28.7	16.2	9.4	22.6	8.7	14.4
U3 - Beta(1, 1)	31.4	15.3	9	22	7.8	14.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.2	5.2	5.9	5.7	4.7	-
U2 - Beta(0.5, 0.5)	6.2	5.4	5.8	5.7	4.6	-
U2 - Beta(0.7, 0.3)	6.4	5.5	5.8	5.7	4.4	-
U2 - Beta(1, 1)	6.4	5.4	5.7	5.7	4.5	-
U3 - Beta(0.3, 0.7)	6.2	5.1	6	5.7	4.7	-
U3 - Beta(0.5, 0.5)	6.3	5.3	5.9	5.7	4.5	-
U3 - Beta(0.7, 0.3)	6.3	5.5	5.8	5.7	4.4	-
U3 - Beta(1, 1)	6.4	5.5	5.7	5.7	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.1	2	2.2	1.4	-
U2 - Beta(0.5, 0.5)	1.4	3.1	2	2.2	1.4	-
U2 - Beta(0.7, 0.3)	1.5	3.2	2	2.2	1.3	-
U2 - Beta(1, 1)	1.5	3.1	2	2.2	1.4	-
U3 - Beta(0.3, 0.7)	1.4	3	2.1	2.2	1.4	-
U3 - Beta(0.5, 0.5)	1.5	3.1	2.1	2.2	1.4	-
U3 - Beta(0.7, 0.3)	1.5	3.2	2	2.2	1.4	-
U3 - Beta(1, 1)	1.5	3.2	2	2.2	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						967.8
U2 - Beta(0.5, 0.5)						968.8
U2 - Beta(0.7, 0.3)						967.1
U2 - Beta(1, 1)						966.2
U3 - Beta(0.3, 0.7)						966
U3 - Beta(0.5, 0.5)						964.8
U3 - Beta(0.7, 0.3)						965.4
U3 - Beta(1, 1)						965.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	41.7	24.7	11	2.7	0.4	19.5
U2 - Beta(0.5, 0.5)	42.8	25.1	9.2	3	0.3	19.6
U2 - Beta(0.7, 0.3)	42.2	25.8	9.4	2.3	0.3	20
U2 - Beta(1, 1)	44.9	24.6	8.9	2.2	0.2	19.2
U3 - Beta(0.3, 0.7)	48.5	24.2	6.2	1.3	0.2	19.6
U3 - Beta(0.5, 0.5)	50.7	22.8	5.7	1.4	0.1	19.3
U3 - Beta(0.7, 0.3)	52.1	21.2	5.6	1.2	0.2	19.7
U3 - Beta(1, 1)	55.7	19.7	3.7	0.9	0.1	19.9
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	6.9	5.9	2.9	1.2	-
U2 - Beta(0.5, 0.5)	10.5	7	5.7	3	1.2	-
U2 - Beta(0.7, 0.3)	10.7	7.1	5.5	2.9	1.2	-
U2 - Beta(1, 1)	10.7	7.1	5.5	2.9	1.2	-
U3 - Beta(0.3, 0.7)	10.5	6.9	5.8	3	1.2	-
U3 - Beta(0.5, 0.5)	10.6	6.9	5.7	2.9	1.2	-
U3 - Beta(0.7, 0.3)	10.6	7.1	5.6	2.9	1.2	-
U3 - Beta(1, 1)	10.7	7.1	5.6	2.9	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.5	2	1.3	0.5	-
U2 - Beta(0.5, 0.5)	3.2	3.5	2	1.3	0.5	-
U2 - Beta(0.7, 0.3)	3.3	3.5	1.9	1.3	0.5	-
U2 - Beta(1, 1)	3.3	3.6	1.9	1.2	0.5	-
U3 - Beta(0.3, 0.7)	3.2	3.5	2	1.3	0.5	-
U3 - Beta(0.5, 0.5)	3.2	3.5	2	1.3	0.5	-
U3 - Beta(0.7, 0.3)	3.2	3.6	1.9	1.3	0.5	-
U3 - Beta(1, 1)	3.3	3.6	2	1.2	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						929.5
U2 - Beta(0.5, 0.5)						928.5
U2 - Beta(0.7, 0.3)						928.6
U2 - Beta(1, 1)						929
U3 - Beta(0.3, 0.7)						929.4
U3 - Beta(0.5, 0.5)						929.4
U3 - Beta(0.7, 0.3)						928.8
U3 - Beta(1, 1)						928.1

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.3	24.7	16.8	18	13.1	0.1
U2 - Beta(0.5, 0.5)	29.9	25	16.3	16.5	12.2	0.1
U2 - Beta(0.7, 0.3)	29.5	27.2	16	15.4	11.8	0.1
U2 - Beta(1, 1)	30.3	28.5	15.8	14.9	10.4	0.1
U3 - Beta(0.3, 0.7)	33.2	30.2	16.1	14.3	6.1	0.1
U3 - Beta(0.5, 0.5)	36.7	29.2	13.2	13.9	6.9	0.1
U3 - Beta(0.7, 0.3)	39.8	29.1	14.3	11.6	5.1	0.1
U3 - Beta(1, 1)	42.7	29.1	14.3	10	3.8	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.8	5.6	5.4	5.8	7.4	-
U2 - Beta(0.5, 0.5)	5.8	5.7	5.5	5.8	7.2	-
U2 - Beta(0.7, 0.3)	5.8	5.8	5.5	5.8	7	-
U2 - Beta(1, 1)	5.8	5.8	5.6	5.8	7.1	-
U3 - Beta(0.3, 0.7)	5.7	5.5	5.5	5.8	7.5	-
U3 - Beta(0.5, 0.5)	5.8	5.8	5.5	5.8	7.2	-
U3 - Beta(0.7, 0.3)	5.9	5.8	5.5	5.8	7	-
U3 - Beta(1, 1)	5.8	5.8	5.5	5.8	7.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.1	1.9	2.6	2.8	-
U2 - Beta(0.5, 0.5)	1.2	3.2	1.9	2.6	2.8	-
U2 - Beta(0.7, 0.3)	1.2	3.2	1.9	2.6	2.7	-
U2 - Beta(1, 1)	1.2	3.2	1.9	2.6	2.7	-
U3 - Beta(0.3, 0.7)	1.2	3.1	1.9	2.6	2.9	-
U3 - Beta(0.5, 0.5)	1.2	3.2	1.9	2.6	2.7	-
U3 - Beta(0.7, 0.3)	1.2	3.2	1.9	2.6	2.7	-
U3 - Beta(1, 1)	1.2	3.2	1.9	2.6	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008.3
U2 - Beta(0.5, 0.5)						1008
U2 - Beta(0.7, 0.3)						1007.4
U2 - Beta(1, 1)						1007.5
U3 - Beta(0.3, 0.7)						1008.7
U3 - Beta(0.5, 0.5)						1007.7
U3 - Beta(0.7, 0.3)						1008.7
U3 - Beta(1, 1)						1008.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.3	51.5	15.1	1.3	0	0.8
U2 - Beta(0.5, 0.5)	30.8	50.8	16.2	1.3	0	0.9
U2 - Beta(0.7, 0.3)	30.9	51.4	14.9	1.9	0	0.9
U2 - Beta(1, 1)	31.3	50.4	16	1.4	0	0.9
U3 - Beta(0.3, 0.7)	33.8	51	13.2	1.1	0	0.9
U3 - Beta(0.5, 0.5)	33.3	52.2	12.6	1	0	0.9
U3 - Beta(0.7, 0.3)	34.6	50.9	12.6	1.1	0	0.8
U3 - Beta(1, 1)	33.8	53	11.4	0.9	0	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	10.1	7	2	0.3	-
U2 - Beta(0.5, 0.5)	10.7	10	6.9	2	0.3	-
U2 - Beta(0.7, 0.3)	10.7	10.2	6.8	1.9	0.3	-
U2 - Beta(1, 1)	10.7	10	7	1.9	0.3	-
U3 - Beta(0.3, 0.7)	10.6	10.1	7	1.9	0.3	-
U3 - Beta(0.5, 0.5)	10.7	10.1	6.9	1.9	0.3	-
U3 - Beta(0.7, 0.3)	10.7	10.1	6.8	1.9	0.3	-
U3 - Beta(1, 1)	10.6	10.1	6.9	2	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	3.5	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.7	3.4	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.7	3.5	1.5	0.6	0.1	-
U2 - Beta(1, 1)	2.7	3.5	1.6	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.7	3.5	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.7	3.5	1.6	0.5	0.1	-
U3 - Beta(0.7, 0.3)	2.7	3.5	1.5	0.6	0.1	-
U3 - Beta(1, 1)	2.7	3.5	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1187.6
U2 - Beta(0.5, 0.5)						1186.5
U2 - Beta(0.7, 0.3)						1187.1
U2 - Beta(1, 1)						1186.3
U3 - Beta(0.3, 0.7)						1187.2
U3 - Beta(0.5, 0.5)						1187.1
U3 - Beta(0.7, 0.3)						1186.1
U3 - Beta(1, 1)						1187.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8	9	10.1	15.4	56.4	1.1
U2 - Beta(0.5, 0.5)	10.4	10.1	9.6	14.4	54.6	0.9
U2 - Beta(0.7, 0.3)	9.5	10.8	10.4	14.9	53.4	1
U2 - Beta(1, 1)	12.2	11.1	8.8	13	53.9	1
U3 - Beta(0.3, 0.7)	12.8	12.7	10.3	12.5	50.6	1.1
U3 - Beta(0.5, 0.5)	15.2	13.9	9.7	11.5	48.7	1
U3 - Beta(0.7, 0.3)	17.5	14.3	10	12.2	45	1
U3 - Beta(1, 1)	18.7	14.1	9.8	11.5	44.8	1.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	3.8	4.7	5.6	12.2	-
U2 - Beta(0.5, 0.5)	3.7	3.8	4.8	5.5	12	-
U2 - Beta(0.7, 0.3)	3.7	4	4.8	5.6	11.8	-
U2 - Beta(1, 1)	3.7	4	4.7	5.5	11.9	-
U3 - Beta(0.3, 0.7)	3.7	3.8	4.7	5.5	12.2	-
U3 - Beta(0.5, 0.5)	3.7	3.9	4.7	5.5	12.1	-
U3 - Beta(0.7, 0.3)	3.7	4	4.8	5.5	11.9	-
U3 - Beta(1, 1)	3.7	4	4.7	5.5	11.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.6	1.5	2.5	3.4	-
U2 - Beta(0.5, 0.5)	0.3	2.7	1.6	2.5	3.4	-
U2 - Beta(0.7, 0.3)	0.4	2.7	1.6	2.5	3.3	-
U2 - Beta(1, 1)	0.3	2.8	1.5	2.5	3.3	-
U3 - Beta(0.3, 0.7)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(0.5, 0.5)	0.3	2.7	1.6	2.5	3.3	-
U3 - Beta(0.7, 0.3)	0.3	2.7	1.6	2.5	3.3	-
U3 - Beta(1, 1)	0.4	2.7	1.6	2.5	3.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1072.8
U2 - Beta(0.5, 0.5)						1070.4
U2 - Beta(0.7, 0.3)						1068.6
U2 - Beta(1, 1)						1069.2
U3 - Beta(0.3, 0.7)						1074.1
U3 - Beta(0.5, 0.5)						1071.4
U3 - Beta(0.7, 0.3)						1069.4
U3 - Beta(1, 1)						1070

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.9	49.2	13.8	0.7	0	17.4
U2 - Beta(0.5, 0.5)	18.7	49.9	13.6	0.9	0	16.9
U2 - Beta(0.7, 0.3)	18.9	49.5	12.9	1.2	0	17.5
U2 - Beta(1, 1)	19.6	50	12.3	0.7	0	17.4
U3 - Beta(0.3, 0.7)	20.6	50	11.7	0.4	0	17.3
U3 - Beta(0.5, 0.5)	20.6	49.8	11.4	0.7	0	17.5
U3 - Beta(0.7, 0.3)	21.2	50.4	10.2	0.7	0	17.5
U3 - Beta(1, 1)	22.7	49.2	10.4	0.5	0	17.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	10.3	6.7	1.7	0.3	-
U2 - Beta(0.5, 0.5)	8.1	10.5	6.4	1.7	0.3	-
U2 - Beta(0.7, 0.3)	8.2	10.7	6.2	1.7	0.3	-
U2 - Beta(1, 1)	8.2	10.6	6.3	1.7	0.3	-
U3 - Beta(0.3, 0.7)	8.2	10.4	6.6	1.7	0.3	-
U3 - Beta(0.5, 0.5)	8.1	10.5	6.5	1.7	0.3	-
U3 - Beta(0.7, 0.3)	8.2	10.6	6.3	1.7	0.3	-
U3 - Beta(1, 1)	8.2	10.6	6.3	1.7	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	4.5	1.9	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.1	4.5	1.8	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.1	4.6	1.8	0.6	0.1	-
U2 - Beta(1, 1)	2.1	4.6	1.8	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.1	4.5	1.9	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.1	4.5	1.9	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.1	4.6	1.8	0.6	0.1	-
U3 - Beta(1, 1)	2.1	4.6	1.8	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						987.5
U2 - Beta(0.5, 0.5)						986.3
U2 - Beta(0.7, 0.3)						985.2
U2 - Beta(1, 1)						985.6
U3 - Beta(0.3, 0.7)						987.1
U3 - Beta(0.5, 0.5)						986.7
U3 - Beta(0.7, 0.3)						985.3
U3 - Beta(1, 1)						985.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.8	47	27.5	8.1	0.8	6.8
U2 - Beta(0.5, 0.5)	11	44.4	27.7	9.2	0.8	6.9
U2 - Beta(0.7, 0.3)	11.4	46.3	25.4	9.3	0.7	6.9
U2 - Beta(1, 1)	13.3	46	25.1	8.1	0.7	6.8
U3 - Beta(0.3, 0.7)	14.9	48.5	23.6	5.6	0.4	7
U3 - Beta(0.5, 0.5)	16	49	20.7	7.2	0.3	6.8
U3 - Beta(0.7, 0.3)	18.7	47.5	19.9	6.9	0.3	6.7
U3 - Beta(1, 1)	22.3	45.9	18.2	6.2	0.3	7.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	9	8.5	4.2	1.8	-
U2 - Beta(0.5, 0.5)	5.3	9.1	8.4	4.1	1.8	-
U2 - Beta(0.7, 0.3)	5.4	9.1	8.4	4.1	1.8	-
U2 - Beta(1, 1)	5.4	9.2	8.3	4.1	1.8	-
U3 - Beta(0.3, 0.7)	5.3	8.9	8.5	4.1	1.9	-
U3 - Beta(0.5, 0.5)	5.3	9	8.4	4.1	1.8	-
U3 - Beta(0.7, 0.3)	5.3	9.1	8.4	4.1	1.8	-
U3 - Beta(1, 1)	5.4	9.2	8.4	4	1.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	4.4	2.6	1.5	0.6	-
U2 - Beta(0.5, 0.5)	1	4.4	2.6	1.5	0.6	-
U2 - Beta(0.7, 0.3)	1.1	4.4	2.6	1.5	0.6	-
U2 - Beta(1, 1)	1.1	4.5	2.6	1.5	0.6	-
U3 - Beta(0.3, 0.7)	1	4.3	2.6	1.5	0.6	-
U3 - Beta(0.5, 0.5)	1	4.4	2.6	1.5	0.6	-
U3 - Beta(0.7, 0.3)	1.1	4.4	2.6	1.6	0.6	-
U3 - Beta(1, 1)	1	4.4	2.6	1.5	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016.9
U2 - Beta(0.5, 0.5)						1014.2
U2 - Beta(0.7, 0.3)						1014.2
U2 - Beta(1, 1)						1015.5
U3 - Beta(0.3, 0.7)						1016.9
U3 - Beta(0.5, 0.5)						1014.6
U3 - Beta(0.7, 0.3)						1014.9
U3 - Beta(1, 1)						1014.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	54.8	0.7	0	0	0	44.5
U2 - Beta(0.5, 0.5)	54.4	1	0	0	0	44.6
U2 - Beta(0.7, 0.3)	54.3	1	0	0	0	44.7
U2 - Beta(1, 1)	54.3	1	0	0	0	44.7
U3 - Beta(0.3, 0.7)	54.3	0.9	0	0	0	44.8
U3 - Beta(0.5, 0.5)	55	1	0	0	0	44
U3 - Beta(0.7, 0.3)	54.7	1	0	0	0	44.3
U3 - Beta(1, 1)	54.9	0.9	0	0	0	44.2
Mean administrations						
U2 - Beta(0.3, 0.7)	20.4	1.9	0.5	0	0	-
U2 - Beta(0.5, 0.5)	20.4	1.9	0.4	0	0	-
U2 - Beta(0.7, 0.3)	20.4	1.8	0.4	0	0	-
U2 - Beta(1, 1)	20.4	1.9	0.4	0	0	-
U3 - Beta(0.3, 0.7)	20.4	1.9	0.4	0	0	-
U3 - Beta(0.5, 0.5)	20.4	1.9	0.4	0	0	-
U3 - Beta(0.7, 0.3)	20.4	1.9	0.4	0	0	-
U3 - Beta(1, 1)	20.4	1.9	0.4	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	0.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.1	0.6	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4.1	0.6	0.1	0	0	-
U2 - Beta(1, 1)	4.1	0.6	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.2	0.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.2	0.6	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4.1	0.6	0.1	0	0	-
U3 - Beta(1, 1)	4.2	0.6	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						990
U2 - Beta(0.5, 0.5)						988.1
U2 - Beta(0.7, 0.3)						986.1
U2 - Beta(1, 1)						987.9
U3 - Beta(0.3, 0.7)						988.3
U3 - Beta(0.5, 0.5)						989.9
U3 - Beta(0.7, 0.3)						988.8
U3 - Beta(1, 1)						989.5
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	24.8	54.3	9.9	1.1	0	9.9
U2 - Beta(0.5, 0.5)	24.4	54.3	9.5	1.3	0	10.5
U2 - Beta(0.7, 0.3)	24.9	53.5	9.9	1.4	0	10.3
U2 - Beta(1, 1)	25.4	52.7	10.6	1.4	0	9.9
U3 - Beta(0.3, 0.7)	26.2	54.7	7.9	1.1	0	10.1
U3 - Beta(0.5, 0.5)	27.9	53	7.8	1	0	10.3
U3 - Beta(0.7, 0.3)	29.7	50.7	7.8	1.3	0	10.5
U3 - Beta(1, 1)	30.3	50.4	7.6	1.3	0	10.4
Mean administrations						
U2 - Beta(0.3, 0.7)	7.9	12.7	5.9	1.3	0.2	-
U2 - Beta(0.5, 0.5)	8.1	12.7	5.8	1.3	0.2	-
U2 - Beta(0.7, 0.3)	8.2	12.6	5.8	1.3	0.2	-
U2 - Beta(1, 1)	8.3	12.6	5.8	1.3	0.2	-
U3 - Beta(0.3, 0.7)	7.9	12.8	5.9	1.3	0.2	-
U3 - Beta(0.5, 0.5)	8.1	12.6	5.9	1.3	0.2	-
U3 - Beta(0.7, 0.3)	8.2	12.6	5.8	1.3	0.2	-
U3 - Beta(1, 1)	8.2	12.6	5.8	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	5.5	1.7	0.5	0	-
U2 - Beta(0.5, 0.5)	2	5.5	1.7	0.5	0	-
U2 - Beta(0.7, 0.3)	2.1	5.4	1.7	0.5	0	-
U2 - Beta(1, 1)	2.1	5.4	1.7	0.4	0	-
U3 - Beta(0.3, 0.7)	2	5.5	1.7	0.5	0	-
U3 - Beta(0.5, 0.5)	2	5.4	1.7	0.5	0	-
U3 - Beta(0.7, 0.3)	2.1	5.4	1.7	0.4	0	-
U3 - Beta(1, 1)	2.1	5.5	1.7	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008
U2 - Beta(0.5, 0.5)						1005.2
U2 - Beta(0.7, 0.3)						1005.5
U2 - Beta(1, 1)						1006.1
U3 - Beta(0.3, 0.7)						1008.4
U3 - Beta(0.5, 0.5)						1006.9
U3 - Beta(0.7, 0.3)						1005.6
U3 - Beta(1, 1)						1004.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.5	3	0	0	0	90.5
U2 - Beta(0.5, 0.5)	6.3	3.3	0	0	0	90.4
U2 - Beta(0.7, 0.3)	6.3	3.3	0	0	0	90.4
U2 - Beta(1, 1)	6.3	3.3	0	0	0	90.4
U3 - Beta(0.3, 0.7)	6.4	3	0	0	0	90.6
U3 - Beta(0.5, 0.5)	6.3	3	0	0	0	90.7
U3 - Beta(0.7, 0.3)	6.7	3.2	0	0	0	90.1
U3 - Beta(1, 1)	6.6	3.2	0	0	0	90.2
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.1	0.4	0	0	-
U2 - Beta(0.5, 0.5)	11.7	2.1	0.4	0	0	-
U2 - Beta(0.7, 0.3)	11.7	2.1	0.4	0	0	-
U2 - Beta(1, 1)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.3, 0.7)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.5, 0.5)	11.7	2.1	0.4	0	0	-
U3 - Beta(0.7, 0.3)	11.7	2.1	0.4	0	0	-
U3 - Beta(1, 1)	11.7	2.1	0.4	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	1.1	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.4	1.1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1.1	0.1	0	0	-
U2 - Beta(1, 1)	3.4	1.1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.4	1.1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.4	1.1	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.4	1.1	0.1	0	0	-
U3 - Beta(1, 1)	3.4	1.1	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						518.5
U2 - Beta(0.5, 0.5)						517.4
U2 - Beta(0.7, 0.3)						517.6
U2 - Beta(1, 1)						517.6
U3 - Beta(0.3, 0.7)						518.7
U3 - Beta(0.5, 0.5)						517.7
U3 - Beta(0.7, 0.3)						518.2
U3 - Beta(1, 1)						518.2
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.6	1.4	4.6	45.4	47.9	0.1
U2 - Beta(0.5, 0.5)	1.1	1.9	4.3	46.8	45.7	0.2
U2 - Beta(0.7, 0.3)	1.5	1.5	5.1	48	43.7	0.2
U2 - Beta(1, 1)	1.4	1.8	4.5	48.2	44	0.1
U3 - Beta(0.3, 0.7)	1.8	2.6	6.2	46.6	42.7	0.1
U3 - Beta(0.5, 0.5)	2.3	2.7	7.1	47	40.8	0.1
U3 - Beta(0.7, 0.3)	3.1	3.9	5.8	49	38.1	0.1
U3 - Beta(1, 1)	3.2	2.9	6.5	48.8	38.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.1	3.6	8.2	11.8	-
U2 - Beta(0.5, 0.5)	3.2	3.2	3.7	8.4	11.5	-
U2 - Beta(0.7, 0.3)	3.3	3.2	3.7	8.5	11.3	-
U2 - Beta(1, 1)	3.2	3.2	3.7	8.4	11.5	-
U3 - Beta(0.3, 0.7)	3.2	3.1	3.7	8.2	11.8	-
U3 - Beta(0.5, 0.5)	3.2	3.2	3.7	8.3	11.6	-
U3 - Beta(0.7, 0.3)	3.2	3.2	3.8	8.5	11.3	-
U3 - Beta(1, 1)	3.3	3.2	3.7	8.4	11.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.2	3.2	3.2	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.1	3.3	3.1	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.2	3.2	3.2	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.3	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1101.2
U2 - Beta(0.5, 0.5)						1099.1
U2 - Beta(0.7, 0.3)						1097.3
U2 - Beta(1, 1)						1099.3
U3 - Beta(0.3, 0.7)						1100.9
U3 - Beta(0.5, 0.5)						1099.3
U3 - Beta(0.7, 0.3)						1097.4
U3 - Beta(1, 1)						1098.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34.3	30.6	29.8	3.4	0	1.9
U2 - Beta(0.5, 0.5)	33.6	31.4	28.9	4.2	0	1.9
U2 - Beta(0.7, 0.3)	34.1	30.9	28.8	4.3	0	1.9
U2 - Beta(1, 1)	34.5	30.3	28.8	4.5	0	1.9
U3 - Beta(0.3, 0.7)	40.1	31.3	23.6	3	0	2
U3 - Beta(0.5, 0.5)	40.9	31.2	23.1	2.9	0	1.9
U3 - Beta(0.7, 0.3)	39.9	31.5	23.1	3.5	0	2
U3 - Beta(1, 1)	41.3	30.5	23.1	3.2	0	1.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	8.3	9.2	2.4	0.4	-
U2 - Beta(0.5, 0.5)	9.5	8.4	9	2.3	0.4	-
U2 - Beta(0.7, 0.3)	9.6	8.6	8.8	2.4	0.4	-
U2 - Beta(1, 1)	9.5	8.5	8.9	2.4	0.4	-
U3 - Beta(0.3, 0.7)	9.3	8.3	9.3	2.3	0.4	-
U3 - Beta(0.5, 0.5)	9.4	8.5	9	2.4	0.4	-
U3 - Beta(0.7, 0.3)	9.5	8.6	8.8	2.3	0.4	-
U3 - Beta(1, 1)	9.6	8.5	8.8	2.3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.7	2.5	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.6	3.7	2.5	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.6	3.7	2.4	0.8	0.1	-
U2 - Beta(1, 1)	2.6	3.7	2.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.7	2.5	0.7	0.1	-
U3 - Beta(0.5, 0.5)	2.6	3.7	2.5	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.6	3.7	2.4	0.8	0.1	-
U3 - Beta(1, 1)	2.6	3.7	2.4	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1106.1
U2 - Beta(0.5, 0.5)						1104.9
U2 - Beta(0.7, 0.3)						1104.2
U2 - Beta(1, 1)						1105.3
U3 - Beta(0.3, 0.7)						1107.1
U3 - Beta(0.5, 0.5)						1105.3
U3 - Beta(0.7, 0.3)						1103.4
U3 - Beta(1, 1)						1104.8

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.7	27.2	26.5	3.4	0	20.2
U2 - Beta(0.5, 0.5)	22	30	24.4	3.8	0	19.8
U2 - Beta(0.7, 0.3)	22.4	29.4	23.5	4.4	0	20.3
U2 - Beta(1, 1)	23.6	27.8	24.7	3.9	0	20
U3 - Beta(0.3, 0.7)	25.6	28	22.9	3.4	0	20.1
U3 - Beta(0.5, 0.5)	26.7	26.9	23	3.1	0	20.3
U3 - Beta(0.7, 0.3)	26.7	28.5	20.4	4.1	0	20.3
U3 - Beta(1, 1)	29.9	27	19.6	3.6	0	19.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	6.8	7.7	2.3	0.3	-
U2 - Beta(0.5, 0.5)	9.8	6.9	7.6	2.3	0.3	-
U2 - Beta(0.7, 0.3)	9.9	6.9	7.5	2.3	0.3	-
U2 - Beta(1, 1)	10	6.9	7.5	2.3	0.3	-
U3 - Beta(0.3, 0.7)	9.8	6.8	7.7	2.3	0.3	-
U3 - Beta(0.5, 0.5)	9.9	7	7.5	2.3	0.3	-
U3 - Beta(0.7, 0.3)	9.9	7	7.4	2.3	0.3	-
U3 - Beta(1, 1)	9.9	6.9	7.4	2.3	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.2	2.3	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.8	3.3	2.2	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.9	3.3	2.2	0.8	0.1	-
U2 - Beta(1, 1)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.8	3.2	2.3	0.8	0.1	-
U3 - Beta(0.5, 0.5)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.9	3.3	2.2	0.8	0.1	-
U3 - Beta(1, 1)	2.9	3.3	2.2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						974.5
U2 - Beta(0.5, 0.5)						973.4
U2 - Beta(0.7, 0.3)						973.4
U2 - Beta(1, 1)						974.1
U3 - Beta(0.3, 0.7)						974.6
U3 - Beta(0.5, 0.5)						971.9
U3 - Beta(0.7, 0.3)						971.8
U3 - Beta(1, 1)						972.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.7	24.3	39.9	6.8	0.3	12
U2 - Beta(0.5, 0.5)	17.4	24.6	39	6.3	0.3	12.4
U2 - Beta(0.7, 0.3)	18	25.3	37.3	6.7	0.3	12.4
U2 - Beta(1, 1)	17.4	25.2	38.1	6.7	0.3	12.3
U3 - Beta(0.3, 0.7)	19.4	25.5	36.3	6.4	0.1	12.3
U3 - Beta(0.5, 0.5)	21.6	25.9	34.4	5.8	0.1	12.2
U3 - Beta(0.7, 0.3)	21.2	26.5	33.4	6.5	0.1	12.3
U3 - Beta(1, 1)	23.4	24.6	34.7	4.9	0.1	12.3
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	6.6	9.2	3.6	1.1	-
U2 - Beta(0.5, 0.5)	7.7	6.7	9.1	3.5	1.1	-
U2 - Beta(0.7, 0.3)	7.7	6.8	9.1	3.5	1	-
U2 - Beta(1, 1)	7.7	6.8	8.9	3.6	1.1	-
U3 - Beta(0.3, 0.7)	7.5	6.7	9.2	3.6	1.1	-
U3 - Beta(0.5, 0.5)	7.7	6.7	9	3.6	1.1	-
U3 - Beta(0.7, 0.3)	7.7	6.8	9	3.6	1	-
U3 - Beta(1, 1)	7.7	6.7	9	3.6	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(0.5, 0.5)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(0.7, 0.3)	2	3.3	2.6	1.1	0.3	-
U2 - Beta(1, 1)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.3, 0.7)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.5, 0.5)	2	3.3	2.6	1.1	0.3	-
U3 - Beta(0.7, 0.3)	2	3.3	2.5	1.1	0.3	-
U3 - Beta(1, 1)	2	3.3	2.6	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1038
U2 - Beta(0.5, 0.5)						1036.7
U2 - Beta(0.7, 0.3)						1037.2
U2 - Beta(1, 1)						1036.3
U3 - Beta(0.3, 0.7)						1038.2
U3 - Beta(0.5, 0.5)						1037
U3 - Beta(0.7, 0.3)						1036.4
U3 - Beta(1, 1)						1036.7
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.7	30.8	4.7	0.1	0	47.7
U2 - Beta(0.5, 0.5)	16.6	30.9	5.1	0	0	47.4
U2 - Beta(0.7, 0.3)	16.5	31.1	4.9	0	0	47.5
U2 - Beta(1, 1)	17	30.8	4.9	0	0	47.3
U3 - Beta(0.3, 0.7)	18.2	30.6	3.9	0	0	47.3
U3 - Beta(0.5, 0.5)	17.5	30.7	3.7	0.1	0	48
U3 - Beta(0.7, 0.3)	17.5	31.6	3.8	0	0	47.1
U3 - Beta(1, 1)	17.7	30.9	4.1	0.1	0	47.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	8.1	3.5	0.7	0.1	-
U2 - Beta(0.5, 0.5)	9.5	8.1	3.5	0.6	0.1	-
U2 - Beta(0.7, 0.3)	9.5	8.1	3.5	0.6	0.1	-
U2 - Beta(1, 1)	9.5	8.1	3.5	0.7	0.1	-
U3 - Beta(0.3, 0.7)	9.4	8.2	3.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	9.4	8.1	3.5	0.6	0.1	-
U3 - Beta(0.7, 0.3)	9.4	8.1	3.5	0.6	0.1	-
U3 - Beta(1, 1)	9.5	8.1	3.5	0.6	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	3.3	1	0.2	0	-
U2 - Beta(0.5, 0.5)	2.6	3.4	1	0.2	0	-
U2 - Beta(0.7, 0.3)	2.6	3.4	1	0.2	0	-
U2 - Beta(1, 1)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.3, 0.7)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.5, 0.5)	2.6	3.4	1	0.2	0	-
U3 - Beta(0.7, 0.3)	2.6	3.4	1	0.2	0	-
U3 - Beta(1, 1)	2.6	3.4	1	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						797.1
U2 - Beta(0.5, 0.5)						797.9
U2 - Beta(0.7, 0.3)						797
U2 - Beta(1, 1)						797.8
U3 - Beta(0.3, 0.7)						797.2
U3 - Beta(0.5, 0.5)						796.9
U3 - Beta(0.7, 0.3)						796.6
U3 - Beta(1, 1)						799.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	72	21.9	3	0.1	0	3
U2 - Beta(0.5, 0.5)	70.8	23.1	2.9	0.2	0	3
U2 - Beta(0.7, 0.3)	72.6	21	3.1	0.3	0	3
U2 - Beta(1, 1)	71.9	22.3	2.7	0.1	0	3
U3 - Beta(0.3, 0.7)	74	20.9	1.9	0.2	0	3
U3 - Beta(0.5, 0.5)	73.1	21.4	2.4	0.1	0	3
U3 - Beta(0.7, 0.3)	73.7	20.6	2.7	0	0	3
U3 - Beta(1, 1)	73	21.8	2.2	0.1	0	2.9
Mean administrations						
U2 - Beta(0.3, 0.7)	18.5	7.3	3.2	0.4	0	-
U2 - Beta(0.5, 0.5)	18.7	7.2	3.1	0.4	0	-
U2 - Beta(0.7, 0.3)	18.7	7.2	3.1	0.4	0	-
U2 - Beta(1, 1)	18.6	7.2	3.2	0.4	0	-
U3 - Beta(0.3, 0.7)	18.6	7.2	3.2	0.4	0	-
U3 - Beta(0.5, 0.5)	18.6	7.3	3.1	0.4	0	-
U3 - Beta(0.7, 0.3)	18.7	7.2	3.1	0.4	0	-
U3 - Beta(1, 1)	18.7	7.3	3.1	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.8	2.4	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	4.9	2.4	0.9	0.1	0	-
U2 - Beta(0.7, 0.3)	4.9	2.4	0.9	0.1	0	-
U2 - Beta(1, 1)	4.8	2.4	0.9	0.1	0	-
U3 - Beta(0.3, 0.7)	4.8	2.4	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	4.8	2.4	0.8	0.1	0	-
U3 - Beta(0.7, 0.3)	4.8	2.4	0.9	0.1	0	-
U3 - Beta(1, 1)	4.9	2.4	0.8	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1171.1
U2 - Beta(0.5, 0.5)						1169.8
U2 - Beta(0.7, 0.3)						1170.2
U2 - Beta(1, 1)						1169.9
U3 - Beta(0.3, 0.7)						1171.3
U3 - Beta(0.5, 0.5)						1169.9
U3 - Beta(0.7, 0.3)						1170.1
U3 - Beta(1, 1)						1169.8

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.3	14.8	15.9	37.7	10.7	1.6
U2 - Beta(0.5, 0.5)	18.7	15.7	17.5	36.2	10.3	1.6
U2 - Beta(0.7, 0.3)	20.3	16.9	15.3	36.1	9.8	1.6
U2 - Beta(1, 1)	21	16.8	14.5	36.5	9.6	1.6
U3 - Beta(0.3, 0.7)	24.5	16.4	14.3	33.8	9.4	1.6
U3 - Beta(0.5, 0.5)	25.5	17.6	13.8	32.2	9.2	1.7
U3 - Beta(0.7, 0.3)	27.3	18.8	13.3	30.9	8	1.7
U3 - Beta(1, 1)	28.2	19.8	12.4	30.2	7.8	1.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	4.5	6.1	8.2	5.7	-
U2 - Beta(0.5, 0.5)	5.1	4.7	6.2	8	5.7	-
U2 - Beta(0.7, 0.3)	5.2	4.8	6.1	8.1	5.6	-
U2 - Beta(1, 1)	5.2	4.8	5.9	8.1	5.6	-
U3 - Beta(0.3, 0.7)	5.2	4.5	6	8.3	5.7	-
U3 - Beta(0.5, 0.5)	5.2	4.7	6.1	8.1	5.6	-
U3 - Beta(0.7, 0.3)	5.2	4.8	6	8.1	5.5	-
U3 - Beta(1, 1)	5.2	4.8	6	8.1	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	2.7	1.9	2.9	1.4	-
U2 - Beta(0.5, 0.5)	1	2.8	2	2.8	1.4	-
U2 - Beta(0.7, 0.3)	1	2.8	1.9	2.8	1.4	-
U2 - Beta(1, 1)	1	2.8	1.9	2.8	1.4	-
U3 - Beta(0.3, 0.7)	1	2.7	1.9	2.9	1.4	-
U3 - Beta(0.5, 0.5)	1	2.8	1.9	2.8	1.4	-
U3 - Beta(0.7, 0.3)	1	2.8	1.9	2.8	1.4	-
U3 - Beta(1, 1)	1	2.8	1.9	2.8	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1091.3
U2 - Beta(0.5, 0.5)						1090
U2 - Beta(0.7, 0.3)						1087.7
U2 - Beta(1, 1)						1089.7
U3 - Beta(0.3, 0.7)						1092.1
U3 - Beta(0.5, 0.5)						1089.3
U3 - Beta(0.7, 0.3)						1088.1
U3 - Beta(1, 1)						1088.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.7	34.1	28.6	24.9	6.4	0.3
U2 - Beta(0.5, 0.5)	5.9	35.8	27.1	23.5	7.4	0.3
U2 - Beta(0.7, 0.3)	6.3	36.6	27.7	22.9	6.2	0.3
U2 - Beta(1, 1)	6.3	36	27.9	23.3	6.2	0.3
U3 - Beta(0.3, 0.7)	8.2	38.5	26.6	21.6	4.8	0.3
U3 - Beta(0.5, 0.5)	8.5	43	24.9	18.9	4.4	0.3
U3 - Beta(0.7, 0.3)	8.7	43.4	23.9	19.2	4.5	0.3
U3 - Beta(1, 1)	9.4	43.5	24.6	18.5	3.7	0.3
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.7	6.4	7.9	5.8	-
U2 - Beta(0.5, 0.5)	4.1	5.8	6.4	7.9	5.7	-
U2 - Beta(0.7, 0.3)	4.1	5.9	6.5	7.8	5.7	-
U2 - Beta(1, 1)	4.1	5.8	6.5	7.8	5.8	-
U3 - Beta(0.3, 0.7)	4.1	5.6	6.4	8	5.8	-
U3 - Beta(0.5, 0.5)	4.1	5.7	6.5	7.9	5.8	-
U3 - Beta(0.7, 0.3)	4.1	5.9	6.4	7.8	5.7	-
U3 - Beta(1, 1)	4.1	5.8	6.5	7.9	5.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.6	2	1.5	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.5	-
U2 - Beta(1, 1)	0.5	2.9	1.7	2	1.5	-
U3 - Beta(0.3, 0.7)	0.5	2.8	1.6	2.1	1.5	-
U3 - Beta(0.5, 0.5)	0.5	2.9	1.6	2	1.5	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.5	-
U3 - Beta(1, 1)	0.5	2.9	1.6	2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1186.8
U2 - Beta(0.5, 0.5)						1185.1
U2 - Beta(0.7, 0.3)						1184.2
U2 - Beta(1, 1)						1186.2
U3 - Beta(0.3, 0.7)						1186.1
U3 - Beta(0.5, 0.5)						1185.5
U3 - Beta(0.7, 0.3)						1183.6
U3 - Beta(1, 1)						1185.2
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.8	13.3	35	29.7	13.4	0.8
U2 - Beta(0.5, 0.5)	8.4	14.5	34.9	28.2	13.2	0.8
U2 - Beta(0.7, 0.3)	9.4	15.7	34.1	29	11	0.8
U2 - Beta(1, 1)	10.2	15.1	36.2	27.6	10.1	0.8
U3 - Beta(0.3, 0.7)	12.8	18.5	34.3	25.6	8.1	0.7
U3 - Beta(0.5, 0.5)	13.5	19.4	33.7	25.8	6.8	0.8
U3 - Beta(0.7, 0.3)	15.6	18.4	34.1	25.1	6	0.8
U3 - Beta(1, 1)	18.8	19.4	33.4	22.5	5.1	0.8
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	4.2	7.7	7.3	6.8	-
U2 - Beta(0.5, 0.5)	4	4.3	7.7	7.3	6.6	-
U2 - Beta(0.7, 0.3)	4	4.4	7.7	7.3	6.4	-
U2 - Beta(1, 1)	4	4.4	7.7	7.3	6.5	-
U3 - Beta(0.3, 0.7)	3.9	4.2	7.7	7.4	6.8	-
U3 - Beta(0.5, 0.5)	4	4.3	7.7	7.3	6.5	-
U3 - Beta(0.7, 0.3)	4	4.4	7.8	7.3	6.4	-
U3 - Beta(1, 1)	4	4.4	7.7	7.3	6.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.7	2.3	-
U2 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.7	2.2	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.8	2.1	-
U2 - Beta(1, 1)	0.5	2.9	2.5	2.7	2.2	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.8	2.3	-
U3 - Beta(0.5, 0.5)	0.4	2.8	2.5	2.7	2.2	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.8	2.1	-
U3 - Beta(1, 1)	0.5	2.9	2.5	2.8	2.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1058.3
U2 - Beta(0.5, 0.5)						1056.7
U2 - Beta(0.7, 0.3)						1055
U2 - Beta(1, 1)						1055.7
U3 - Beta(0.3, 0.7)						1057.3
U3 - Beta(0.5, 0.5)						1054.9
U3 - Beta(0.7, 0.3)						1055
U3 - Beta(1, 1)						1054.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.5	6.5	14.3	41.7	29.9	0.1
U2 - Beta(0.5, 0.5)	7.9	6.8	13.3	42	29.8	0.2
U2 - Beta(0.7, 0.3)	9.8	6.7	12.7	41.7	29	0.1
U2 - Beta(1, 1)	10.7	8	12.9	40.4	27.9	0.1
U3 - Beta(0.3, 0.7)	11	8.5	14.3	39.7	26.3	0.2
U3 - Beta(0.5, 0.5)	13.5	9	12.7	39.3	25.4	0.1
U3 - Beta(0.7, 0.3)	15.7	8.2	13.5	40.2	22.3	0.1
U3 - Beta(1, 1)	16.7	8.8	12.4	39.6	22.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.5	5.2	8.1	9.7	-
U2 - Beta(0.5, 0.5)	3.5	3.6	5.3	8.1	9.5	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.3	8.1	9.3	-
U2 - Beta(1, 1)	3.5	3.7	5.2	8.2	9.4	-
U3 - Beta(0.3, 0.7)	3.4	3.6	5.3	8	9.7	-
U3 - Beta(0.5, 0.5)	3.5	3.6	5.3	8.2	9.5	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.3	8.2	9.3	-
U3 - Beta(1, 1)	3.5	3.7	5.2	8.2	9.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.6	3.1	2.3	-
U2 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.4	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.6	3.1	2.4	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.6	3.1	2.3	-
U3 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1108.8
U2 - Beta(0.5, 0.5)						1105.4
U2 - Beta(0.7, 0.3)						1104.3
U2 - Beta(1, 1)						1106.5
U3 - Beta(0.3, 0.7)						1109.1
U3 - Beta(0.5, 0.5)						1106
U3 - Beta(0.7, 0.3)						1104.3
U3 - Beta(1, 1)						1106.8
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.3	13.5	48.6	15.7	4.9	5
U2 - Beta(0.5, 0.5)	13.3	13	48.7	15.6	4.2	5.2
U2 - Beta(0.7, 0.3)	13.9	12.6	48.5	16.1	3.9	5
U2 - Beta(1, 1)	15.7	12	47.5	16.4	3.3	5.1
U3 - Beta(0.3, 0.7)	15.9	16.3	46.1	13.6	3	5.1
U3 - Beta(0.5, 0.5)	17.7	14.5	46.3	14.5	2.2	4.8
U3 - Beta(0.7, 0.3)	22.4	14.2	42.9	13.3	2.3	4.9
U3 - Beta(1, 1)	26.5	12.6	40.7	13.3	2	4.9
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	5.8	9.9	5.3	3	-
U2 - Beta(0.5, 0.5)	5.2	5.9	9.9	5.3	2.8	-
U2 - Beta(0.7, 0.3)	5.3	5.9	9.9	5.2	2.8	-
U2 - Beta(1, 1)	5.3	6	9.8	5.2	2.9	-
U3 - Beta(0.3, 0.7)	5.1	5.8	10	5.3	2.9	-
U3 - Beta(0.5, 0.5)	5.1	5.8	10	5.3	2.9	-
U3 - Beta(0.7, 0.3)	5.3	6	9.9	5.1	2.8	-
U3 - Beta(1, 1)	5.3	5.9	9.8	5.2	2.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.3	3.1	1.9	0.9	-
U2 - Beta(0.5, 0.5)	1	3.3	3.1	1.9	0.9	-
U2 - Beta(0.7, 0.3)	1	3.3	3.1	1.8	0.9	-
U2 - Beta(1, 1)	1.1	3.4	3.1	1.8	0.9	-
U3 - Beta(0.3, 0.7)	1	3.3	3.2	1.8	0.9	-
U3 - Beta(0.5, 0.5)	1	3.3	3.2	1.9	0.9	-
U3 - Beta(0.7, 0.3)	1	3.4	3.1	1.8	0.9	-
U3 - Beta(1, 1)	1.1	3.3	3.1	1.8	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1042.5
U2 - Beta(0.5, 0.5)						1039.3
U2 - Beta(0.7, 0.3)						1037.9
U2 - Beta(1, 1)						1038.6
U3 - Beta(0.3, 0.7)						1043.1
U3 - Beta(0.5, 0.5)						1040.2
U3 - Beta(0.7, 0.3)						1039.2
U3 - Beta(1, 1)						1038

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.4	40.6	34.4	9	3.4	0.2
U2 - Beta(0.5, 0.5)	11.8	40.7	34	10.4	2.9	0.2
U2 - Beta(0.7, 0.3)	12.4	40.7	33.7	10	3	0.2
U2 - Beta(1, 1)	13.5	40.4	33.3	9.8	2.8	0.2
U3 - Beta(0.3, 0.7)	14.7	44.7	29.3	9	2.1	0.2
U3 - Beta(0.5, 0.5)	15.6	47.5	27.1	8	1.6	0.2
U3 - Beta(0.7, 0.3)	17.6	46.3	25.7	8.5	1.7	0.2
U3 - Beta(1, 1)	18	48.2	24.5	7.7	1.4	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	8.4	9.3	5.1	2.7	-
U2 - Beta(0.5, 0.5)	4.6	8.6	9.1	5	2.7	-
U2 - Beta(0.7, 0.3)	4.6	8.7	9.1	5	2.6	-
U2 - Beta(1, 1)	4.5	8.7	9.1	5	2.7	-
U3 - Beta(0.3, 0.7)	4.5	8.4	9.2	5.1	2.8	-
U3 - Beta(0.5, 0.5)	4.5	8.6	9	5.1	2.7	-
U3 - Beta(0.7, 0.3)	4.6	8.8	8.9	5	2.7	-
U3 - Beta(1, 1)	4.5	8.8	8.9	5.1	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.7	2.3	1.5	0.7	-
U2 - Beta(0.5, 0.5)	0.6	3.7	2.3	1.5	0.7	-
U2 - Beta(0.7, 0.3)	0.7	3.8	2.3	1.5	0.7	-
U2 - Beta(1, 1)	0.6	3.8	2.3	1.5	0.7	-
U3 - Beta(0.3, 0.7)	0.6	3.7	2.3	1.5	0.8	-
U3 - Beta(0.5, 0.5)	0.6	3.7	2.2	1.5	0.7	-
U3 - Beta(0.7, 0.3)	0.7	3.8	2.2	1.4	0.7	-
U3 - Beta(1, 1)	0.6	3.8	2.2	1.5	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1166.5
U2 - Beta(0.5, 0.5)						1165.7
U2 - Beta(0.7, 0.3)						1164.6
U2 - Beta(1, 1)						1164.9
U3 - Beta(0.3, 0.7)						1164.8
U3 - Beta(0.5, 0.5)						1165
U3 - Beta(0.7, 0.3)						1163.9
U3 - Beta(1, 1)						1165.3
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.6	37.6	13.6	4.8	1.3	27.1
U2 - Beta(0.5, 0.5)	17.2	37.4	12.9	4.5	1	27
U2 - Beta(0.7, 0.3)	18.1	36.3	12.9	4.5	1	27.2
U2 - Beta(1, 1)	19	36.9	12	4.2	0.6	27.3
U3 - Beta(0.3, 0.7)	18	37.5	11.8	3.8	0.9	28
U3 - Beta(0.5, 0.5)	20.2	37.1	10.7	4.1	0.8	27.1
U3 - Beta(0.7, 0.3)	21.9	36.5	10.1	3.5	0.7	27.3
U3 - Beta(1, 1)	23.5	36.6	8.4	3.5	0.8	27.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	8.2	5.5	2.3	1	-
U2 - Beta(0.5, 0.5)	8.2	8.5	5.2	2.3	0.9	-
U2 - Beta(0.7, 0.3)	8.3	8.5	5.2	2.3	0.9	-
U2 - Beta(1, 1)	8.3	8.4	5.2	2.3	1	-
U3 - Beta(0.3, 0.7)	8.2	8.3	5.4	2.4	0.9	-
U3 - Beta(0.5, 0.5)	8.3	8.3	5.3	2.4	1	-
U3 - Beta(0.7, 0.3)	8.3	8.5	5.2	2.4	1	-
U3 - Beta(1, 1)	8.3	8.4	5.2	2.3	1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.9	1.8	0.9	0.3	-
U2 - Beta(0.5, 0.5)	2.1	4	1.7	0.9	0.3	-
U2 - Beta(0.7, 0.3)	2.1	4	1.7	0.9	0.3	-
U2 - Beta(1, 1)	2.2	4	1.7	0.9	0.3	-
U3 - Beta(0.3, 0.7)	2.1	4	1.7	0.9	0.3	-
U3 - Beta(0.5, 0.5)	2.2	3.9	1.7	0.9	0.3	-
U3 - Beta(0.7, 0.3)	2.2	4	1.6	0.9	0.3	-
U3 - Beta(1, 1)	2.2	4	1.7	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						891.2
U2 - Beta(0.5, 0.5)						890.1
U2 - Beta(0.7, 0.3)						891
U2 - Beta(1, 1)						890.8
U3 - Beta(0.3, 0.7)						889.5
U3 - Beta(0.5, 0.5)						890.9
U3 - Beta(0.7, 0.3)						892.5
U3 - Beta(1, 1)						891.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.3	47.9	11.7	1.6	0	17.5
U2 - Beta(0.5, 0.5)	21.1	46.4	12.1	2.1	0.2	18.1
U2 - Beta(0.7, 0.3)	22.4	46	11.4	2.2	0.2	17.8
U2 - Beta(1, 1)	22.1	46.5	12	1.7	0	17.7
U3 - Beta(0.3, 0.7)	24	47	9.9	1.3	0	17.8
U3 - Beta(0.5, 0.5)	26.7	45	9	1.7	0	17.6
U3 - Beta(0.7, 0.3)	27.7	44.6	8.2	1.6	0	17.9
U3 - Beta(1, 1)	28.4	44.8	8.5	1	0	17.3
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	10.5	6.1	2	0.7	-
U2 - Beta(0.5, 0.5)	7.7	10.6	5.9	1.9	0.7	-
U2 - Beta(0.7, 0.3)	7.8	10.7	5.8	1.9	0.7	-
U2 - Beta(1, 1)	7.8	10.7	5.8	1.9	0.7	-
U3 - Beta(0.3, 0.7)	7.6	10.6	6	2	0.7	-
U3 - Beta(0.5, 0.5)	7.7	10.7	5.8	1.9	0.7	-
U3 - Beta(0.7, 0.3)	7.7	10.6	5.8	1.9	0.7	-
U3 - Beta(1, 1)	7.8	10.7	5.7	1.9	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	4.9	1.9	0.8	0.2	-
U2 - Beta(0.5, 0.5)	1.9	4.9	1.8	0.8	0.2	-
U2 - Beta(0.7, 0.3)	1.9	5	1.8	0.8	0.2	-
U2 - Beta(1, 1)	1.9	5	1.8	0.8	0.2	-
U3 - Beta(0.3, 0.7)	1.9	4.9	1.9	0.8	0.2	-
U3 - Beta(0.5, 0.5)	1.9	5	1.8	0.8	0.2	-
U3 - Beta(0.7, 0.3)	1.9	5	1.8	0.8	0.2	-
U3 - Beta(1, 1)	1.9	5	1.8	0.8	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						942.4
U2 - Beta(0.5, 0.5)						940.5
U2 - Beta(0.7, 0.3)						940.1
U2 - Beta(1, 1)						937.3
U3 - Beta(0.3, 0.7)						941.3
U3 - Beta(0.5, 0.5)						940.7
U3 - Beta(0.7, 0.3)						938.1
U3 - Beta(1, 1)						938.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	7.7	1.4	0.5	0.1	76.5
U2 - Beta(0.5, 0.5)	16	7.6	1.6	0.2	0.1	74.5
U2 - Beta(0.7, 0.3)	15.3	9.3	1.2	0.3	0.1	73.8
U2 - Beta(1, 1)	15.7	8.9	1	0.4	0.1	73.9
U3 - Beta(0.3, 0.7)	13.7	7.9	1	0.1	0	77.3
U3 - Beta(0.5, 0.5)	14.9	9.3	1.3	0.1	0	74.4
U3 - Beta(0.7, 0.3)	15.2	8.6	1.2	0.1	0	74.9
U3 - Beta(1, 1)	14.5	9.3	1	0.1	0	75.1
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	5.2	1.9	0.8	0.1	-
U2 - Beta(0.5, 0.5)	11.1	5.3	1.9	1	0.2	-
U2 - Beta(0.7, 0.3)	11.2	5.3	2	1	0.2	-
U2 - Beta(1, 1)	11.1	5.3	2	1	0.2	-
U3 - Beta(0.3, 0.7)	11.2	5.2	1.8	0.7	0.1	-
U3 - Beta(0.5, 0.5)	11.3	5.2	1.9	0.9	0.2	-
U3 - Beta(0.7, 0.3)	11.3	5.2	1.9	1	0.2	-
U3 - Beta(1, 1)	11.3	5.2	1.8	0.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.7	0.7	0.4	0.1	-
U2 - Beta(0.5, 0.5)	3	2.8	0.6	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3	2.8	0.7	0.4	0.1	-
U2 - Beta(1, 1)	3	2.8	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.1	2.7	0.6	0.4	0.1	-
U3 - Beta(0.5, 0.5)	3.1	2.7	0.6	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.1	2.8	0.6	0.4	0.1	-
U3 - Beta(1, 1)	3	2.8	0.6	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						677.1
U2 - Beta(0.5, 0.5)						688.5
U2 - Beta(0.7, 0.3)						691.5
U2 - Beta(1, 1)						690
U3 - Beta(0.3, 0.7)						672.5
U3 - Beta(0.5, 0.5)						687.8
U3 - Beta(0.7, 0.3)						688.3
U3 - Beta(1, 1)						686.7
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.8	31.1	34.6	3.8	0.6	3.1
U2 - Beta(0.5, 0.5)	27.1	20.1	37.4	11.4	1.1	2.9
U2 - Beta(0.7, 0.3)	27.5	20.3	35.3	12.2	2	2.7
U2 - Beta(1, 1)	26.8	19.5	38.5	11.6	1	2.6
U3 - Beta(0.3, 0.7)	36	33.7	24.3	2.7	0.1	3.2
U3 - Beta(0.5, 0.5)	30.9	20.6	34.7	10	0.9	2.9
U3 - Beta(0.7, 0.3)	30.7	20.6	32.4	12	1.5	2.8
U3 - Beta(1, 1)	31.6	20.1	35	9.9	0.6	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	12.3	9.8	6.6	0.8	0.1	-
U2 - Beta(0.5, 0.5)	9.6	7.5	8.8	3.1	0.5	-
U2 - Beta(0.7, 0.3)	9.7	7.1	7.5	4.2	1.1	-
U2 - Beta(1, 1)	9.4	7.6	8.9	3.1	0.5	-
U3 - Beta(0.3, 0.7)	13.9	10	5.1	0.6	0	-
U3 - Beta(0.5, 0.5)	10.3	7.7	8.4	2.8	0.4	-
U3 - Beta(0.7, 0.3)	10	7.3	7.6	4	0.8	-
U3 - Beta(1, 1)	10.2	7.8	8.5	2.7	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	4.1	2	0.2	0	-
U2 - Beta(0.5, 0.5)	2.4	3.4	2.4	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.3	2	1.2	0.3	-
U2 - Beta(1, 1)	2.3	3.4	2.5	0.9	0.1	-
U3 - Beta(0.3, 0.7)	4.3	4.2	1.5	0.2	0	-
U3 - Beta(0.5, 0.5)	2.7	3.5	2.3	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.6	3.3	2.1	1.1	0.2	-
U3 - Beta(1, 1)	2.7	3.5	2.3	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1077.4
U2 - Beta(0.5, 0.5)						1115.2
U2 - Beta(0.7, 0.3)						1121.9
U2 - Beta(1, 1)						1115.6
U3 - Beta(0.3, 0.7)						1065.9
U3 - Beta(0.5, 0.5)						1110.9
U3 - Beta(0.7, 0.3)						1116.6
U3 - Beta(1, 1)						1109.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.9	64.9	16.9	3.4	0	8.9
U2 - Beta(0.5, 0.5)	8.2	47.6	24.3	9.6	1.4	8.9
U2 - Beta(0.7, 0.3)	7.8	49.7	22.7	9.4	1.6	8.8
U2 - Beta(1, 1)	8.1	46.6	26.3	8.4	1.4	9.2
U3 - Beta(0.3, 0.7)	7.2	68.8	12.5	2.3	0	9.2
U3 - Beta(0.5, 0.5)	9.3	53.2	20.5	6.8	1	9.2
U3 - Beta(0.7, 0.3)	9.6	50.8	21	8	1.3	9.3
U3 - Beta(1, 1)	8.4	53.7	20.8	7	0.9	9.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	16.5	3.7	0.7	0.1	-
U2 - Beta(0.5, 0.5)	6.8	11.7	6.5	3	0.7	-
U2 - Beta(0.7, 0.3)	6.7	10.3	6.5	4	1.3	-
U2 - Beta(1, 1)	6.7	11.7	6.6	3	0.7	-
U3 - Beta(0.3, 0.7)	8.2	17	2.9	0.5	0	-
U3 - Beta(0.5, 0.5)	6.9	12.5	6.1	2.7	0.5	-
U3 - Beta(0.7, 0.3)	6.8	11	6.4	3.6	0.9	-
U3 - Beta(1, 1)	6.8	12.7	6	2.7	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	6.3	1.2	0.3	0	-
U2 - Beta(0.5, 0.5)	1.3	4.8	1.9	1	0.2	-
U2 - Beta(0.7, 0.3)	1.3	4.4	1.8	1.3	0.4	-
U2 - Beta(1, 1)	1.2	4.8	2	1	0.2	-
U3 - Beta(0.3, 0.7)	2	6.5	0.9	0.2	0	-
U3 - Beta(0.5, 0.5)	1.3	5.1	1.8	0.9	0.2	-
U3 - Beta(0.7, 0.3)	1.3	4.6	1.9	1.2	0.3	-
U3 - Beta(1, 1)	1.3	5.1	1.8	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1053.8
U2 - Beta(0.5, 0.5)						1069.8
U2 - Beta(0.7, 0.3)						1075.4
U2 - Beta(1, 1)						1068.8
U3 - Beta(0.3, 0.7)						1049.6
U3 - Beta(0.5, 0.5)						1066.2
U3 - Beta(0.7, 0.3)						1071.7
U3 - Beta(1, 1)						1065.3

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	5.9	31.9	53	2.7	1.9
U2 - Beta(0.5, 0.5)	6.3	5.7	19.3	54.2	12.8	1.7
U2 - Beta(0.7, 0.3)	7.6	6.9	20.2	50	13.6	1.7
U2 - Beta(1, 1)	6.8	5.8	18.8	54.5	12.5	1.6
U3 - Beta(0.3, 0.7)	5.9	7.6	33.9	48.9	1.9	1.8
U3 - Beta(0.5, 0.5)	8.4	7.5	18.9	53.5	10.1	1.6
U3 - Beta(0.7, 0.3)	9	7	20.4	49.1	12.8	1.7
U3 - Beta(1, 1)	8.8	7.4	18.3	55.3	8.5	1.7
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	5.5	10.3	8.9	0.6	-
U2 - Beta(0.5, 0.5)	4.2	4.7	7.1	10.8	3	-
U2 - Beta(0.7, 0.3)	4.3	4.7	6.9	9.7	4.3	-
U2 - Beta(1, 1)	4.2	4.8	7	10.8	3	-
U3 - Beta(0.3, 0.7)	4.7	5.9	10.4	8.3	0.4	-
U3 - Beta(0.5, 0.5)	4.3	4.8	7.2	10.8	2.7	-
U3 - Beta(0.7, 0.3)	4.5	5	6.9	9.7	3.7	-
U3 - Beta(1, 1)	4.6	5	7	10.7	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.2	3.7	2.9	0.2	-
U2 - Beta(0.5, 0.5)	0.5	3.1	2.1	3.4	0.8	-
U2 - Beta(0.7, 0.3)	0.5	3.1	2	3.1	1.1	-
U2 - Beta(1, 1)	0.5	3.1	2.1	3.4	0.9	-
U3 - Beta(0.3, 0.7)	0.8	3.3	3.7	2.7	0.1	-
U3 - Beta(0.5, 0.5)	0.6	3.1	2.1	3.4	0.7	-
U3 - Beta(0.7, 0.3)	0.6	3.2	2	3.1	1	-
U3 - Beta(1, 1)	0.7	3.2	2	3.4	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1056.9
U2 - Beta(0.5, 0.5)						1098.5
U2 - Beta(0.7, 0.3)						1102.2
U2 - Beta(1, 1)						1098
U3 - Beta(0.3, 0.7)						1050.1
U3 - Beta(0.5, 0.5)						1093.9
U3 - Beta(0.7, 0.3)						1095.9
U3 - Beta(1, 1)						1091.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.2	45.1	27.3	19.4	5.4	1.6
U2 - Beta(0.5, 0.5)	5.4	33.8	26.1	22.9	10.3	1.5
U2 - Beta(0.7, 0.3)	5.7	36.2	27.4	20.8	8.4	1.5
U2 - Beta(1, 1)	6.2	36.7	25.6	20.8	9.2	1.5
U3 - Beta(0.3, 0.7)	2.3	52.2	24.9	15	4	1.6
U3 - Beta(0.5, 0.5)	7.5	42.1	23.3	19	6.6	1.5
U3 - Beta(0.7, 0.3)	7.2	43.8	24.8	16.5	6.2	1.5
U3 - Beta(1, 1)	9.8	44.8	22.3	16.4	5.2	1.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	14	6.9	3.6	1	-
U2 - Beta(0.5, 0.5)	4	7.5	7.2	7.1	3.9	-
U2 - Beta(0.7, 0.3)	4	6.8	7.1	7.4	4.5	-
U2 - Beta(1, 1)	3.9	7.5	7.4	7.1	3.8	-
U3 - Beta(0.3, 0.7)	4.5	14.8	6.6	3.1	0.7	-
U3 - Beta(0.5, 0.5)	4.1	8.2	7.3	6.9	3.3	-
U3 - Beta(0.7, 0.3)	4.1	7.6	7.1	7	3.8	-
U3 - Beta(1, 1)	4.1	8.6	7.3	6.7	3.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	6.1	2.6	1.3	0.3	-
U2 - Beta(0.5, 0.5)	0.3	3.9	2.3	2.7	1.4	-
U2 - Beta(0.7, 0.3)	0.4	3.8	2.2	2.8	1.5	-
U2 - Beta(1, 1)	0.3	3.9	2.4	2.7	1.4	-
U3 - Beta(0.3, 0.7)	0.7	6.5	2.5	1.2	0.3	-
U3 - Beta(0.5, 0.5)	0.4	4.2	2.4	2.6	1.2	-
U3 - Beta(0.7, 0.3)	0.4	4	2.3	2.6	1.4	-
U3 - Beta(1, 1)	0.4	4.4	2.4	2.5	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.1
U2 - Beta(0.5, 0.5)						1047.7
U2 - Beta(0.7, 0.3)						1051
U2 - Beta(1, 1)						1048.6
U3 - Beta(0.3, 0.7)						1030.4
U3 - Beta(0.5, 0.5)						1046.3
U3 - Beta(0.7, 0.3)						1048.3
U3 - Beta(1, 1)						1044.6

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	69	11	2.4	0.4	0	18.2
U2 - Beta(0.5, 0.5)	65.2	13.5	3.6	1.1	0.2	16.4
U2 - Beta(0.7, 0.3)	63.4	14.6	3.8	1.3	0.2	16.7
U2 - Beta(1, 1)	64.5	14.2	3.6	1.1	0.2	16.4
U3 - Beta(0.3, 0.7)	70	9.9	1.9	0.2	0	18
U3 - Beta(0.5, 0.5)	67.4	12.5	3	0.9	0.2	16
U3 - Beta(0.7, 0.3)	66.5	13.5	3	0.6	0.1	16.3
U3 - Beta(1, 1)	66.4	13.3	2.7	0.8	0.1	16.7
Mean administrations						
U2 - Beta(0.3, 0.7)	21.8	4	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	18.3	5.7	2.2	0.8	0.2	-
U2 - Beta(0.7, 0.3)	17.8	6	2.1	1.1	0.3	-
U2 - Beta(1, 1)	18.1	5.9	2.2	0.8	0.2	-
U3 - Beta(0.3, 0.7)	22.3	3.8	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	18.9	5.6	2	0.7	0.2	-
U3 - Beta(0.7, 0.3)	18.2	5.8	2.1	1	0.2	-
U3 - Beta(1, 1)	18.9	5.7	1.8	0.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6	1.7	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	4.7	2.3	0.7	0.3	0.1	-
U2 - Beta(0.7, 0.3)	4.6	2.4	0.6	0.4	0.1	-
U2 - Beta(1, 1)	4.6	2.4	0.7	0.3	0.1	-
U3 - Beta(0.3, 0.7)	6.2	1.6	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.9	2.3	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	4.6	2.4	0.6	0.3	0.1	-
U3 - Beta(1, 1)	4.9	2.4	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1035.2
U2 - Beta(0.5, 0.5)						1058.4
U2 - Beta(0.7, 0.3)						1063.9
U2 - Beta(1, 1)						1059
U3 - Beta(0.3, 0.7)						1035.1
U3 - Beta(0.5, 0.5)						1059.8
U3 - Beta(0.7, 0.3)						1059
U3 - Beta(1, 1)						1056.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.7	70.9	8.2	0.7	0.1	3.4
U2 - Beta(0.5, 0.5)	12.2	51.6	23.5	8	1.8	2.9
U2 - Beta(0.7, 0.3)	13	47.4	22.2	12.5	2.2	2.7
U2 - Beta(1, 1)	12.3	51.4	23	8.1	2	3.2
U3 - Beta(0.3, 0.7)	19.9	70.9	5.2	0.6	0	3.4
U3 - Beta(0.5, 0.5)	14	55.1	20.1	6.8	1.2	2.8
U3 - Beta(0.7, 0.3)	14.2	48.8	22.3	10.6	1.1	3
U3 - Beta(1, 1)	14.2	56	20	5.9	0.8	3.1
Mean administrations						
U2 - Beta(0.3, 0.7)	11.6	16	1.7	0.2	0	-
U2 - Beta(0.5, 0.5)	7.9	13.5	5.7	2	0.4	-
U2 - Beta(0.7, 0.3)	7.7	10.1	6.2	4.2	1.4	-
U2 - Beta(1, 1)	7.9	13.6	5.5	2.1	0.5	-
U3 - Beta(0.3, 0.7)	12.1	16	1.2	0.1	0	-
U3 - Beta(0.5, 0.5)	8.3	14.2	5	1.8	0.3	-
U3 - Beta(0.7, 0.3)	7.9	11.5	6.1	3.4	0.7	-
U3 - Beta(1, 1)	8.3	14.3	5	1.7	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	5.5	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	1.6	4.9	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	1.5	3.9	1.6	1.2	0.4	-
U2 - Beta(1, 1)	1.6	4.9	1.6	0.6	0.1	-
U3 - Beta(0.3, 0.7)	3.4	5.5	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	1.8	5.1	1.3	0.6	0.1	-
U3 - Beta(0.7, 0.3)	1.6	4.3	1.7	1	0.2	-
U3 - Beta(1, 1)	1.7	5.2	1.3	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1118.9
U2 - Beta(0.5, 0.5)						1145.8
U2 - Beta(0.7, 0.3)						1153.2
U2 - Beta(1, 1)						1145.4
U3 - Beta(0.3, 0.7)						1114.9
U3 - Beta(0.5, 0.5)						1145
U3 - Beta(0.7, 0.3)						1148.2
U3 - Beta(1, 1)						1142
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.1	8.5	2.6	0.7	0.1	62
U2 - Beta(0.5, 0.5)	28.3	8.9	1.6	0.3	0.2	60.7
U2 - Beta(0.7, 0.3)	29.1	8.4	1.9	0.3	0.2	60.1
U2 - Beta(1, 1)	28.6	9.3	1.6	0.4	0.2	59.9
U3 - Beta(0.3, 0.7)	27.3	7.5	2	0.7	0.1	62.4
U3 - Beta(0.5, 0.5)	29.2	8.5	1.3	0.3	0.2	60.5
U3 - Beta(0.7, 0.3)	28.9	8.7	1.2	0.4	0.2	60.6
U3 - Beta(1, 1)	29.3	8.2	1.1	0.4	0.2	60.8
Mean administrations						
U2 - Beta(0.3, 0.7)	14	5.1	1.5	0.4	0	-
U2 - Beta(0.5, 0.5)	13.3	5.4	1.8	0.8	0.2	-
U2 - Beta(0.7, 0.3)	13.4	5.3	1.7	0.8	0.2	-
U2 - Beta(1, 1)	13.4	5.4	1.7	0.8	0.2	-
U3 - Beta(0.3, 0.7)	14.2	5.1	1.3	0.3	0	-
U3 - Beta(0.5, 0.5)	13.6	5.2	1.6	0.7	0.1	-
U3 - Beta(0.7, 0.3)	13.4	5.3	1.6	0.8	0.1	-
U3 - Beta(1, 1)	13.6	5.3	1.6	0.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.9	2.5	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.6	2.6	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.7	2.6	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.6	2.6	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	4	2.5	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	3.7	2.6	0.5	0.3	0	-
U3 - Beta(0.7, 0.3)	3.6	2.6	0.5	0.3	0	-
U3 - Beta(1, 1)	3.7	2.6	0.5	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						760.3
U2 - Beta(0.5, 0.5)						774.6
U2 - Beta(0.7, 0.3)						779.2
U2 - Beta(1, 1)						776.9
U3 - Beta(0.3, 0.7)						756.6
U3 - Beta(0.5, 0.5)						774.9
U3 - Beta(0.7, 0.3)						772.8
U3 - Beta(1, 1)						773.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8	11	26	22.3	15.1	17.6
U2 - Beta(0.5, 0.5)	12.3	12.1	20.8	16.4	22.1	16.3
U2 - Beta(0.7, 0.3)	11.7	12.3	21.6	16.2	21.4	16.8
U2 - Beta(1, 1)	12.2	11.3	21.7	16	22.1	16.7
U3 - Beta(0.3, 0.7)	13.2	14	23.3	20.5	11	18
U3 - Beta(0.5, 0.5)	15.9	14.5	19.9	15.2	17.8	16.7
U3 - Beta(0.7, 0.3)	17.8	14	18.9	15.6	17.1	16.6
U3 - Beta(1, 1)	20.4	14.8	17.5	13.8	16.6	16.9
Mean administrations						
U2 - Beta(0.3, 0.7)	7	6.3	7.3	5	2.2	-
U2 - Beta(0.5, 0.5)	6.7	5.9	6	5.1	4.2	-
U2 - Beta(0.7, 0.3)	6.6	5.9	5.8	5	4.6	-
U2 - Beta(1, 1)	6.6	6	6	5.2	4.2	-
U3 - Beta(0.3, 0.7)	7.4	6.4	7.3	4.9	1.7	-
U3 - Beta(0.5, 0.5)	6.9	6	6	5.1	3.9	-
U3 - Beta(0.7, 0.3)	7	6	5.8	5.1	4.1	-
U3 - Beta(1, 1)	7.2	6.2	5.6	5	3.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	3.4	2.7	2	0.7	-
U2 - Beta(0.5, 0.5)	1.5	3.4	2	2	1.3	-
U2 - Beta(0.7, 0.3)	1.4	3.4	1.9	2	1.4	-
U2 - Beta(1, 1)	1.4	3.4	2	2	1.3	-
U3 - Beta(0.3, 0.7)	1.8	3.4	2.7	2	0.6	-
U3 - Beta(0.5, 0.5)	1.6	3.4	2	2	1.2	-
U3 - Beta(0.7, 0.3)	1.6	3.5	1.9	2	1.2	-
U3 - Beta(1, 1)	1.7	3.5	1.9	2	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						954.8
U2 - Beta(0.5, 0.5)						978
U2 - Beta(0.7, 0.3)						981
U2 - Beta(1, 1)						977.4
U3 - Beta(0.3, 0.7)						950
U3 - Beta(0.5, 0.5)						974.7
U3 - Beta(0.7, 0.3)						974.9
U3 - Beta(1, 1)						972

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.6	38.7	8.5	1.8	0	44.4
U2 - Beta(0.5, 0.5)	7.6	32.5	13.5	3.6	0.9	41.9
U2 - Beta(0.7, 0.3)	8.1	32.4	13.4	4.6	1.2	40.3
U2 - Beta(1, 1)	8	31.4	14	3.9	0.9	41.8
U3 - Beta(0.3, 0.7)	7	39.7	7.6	1.1	0	44.6
U3 - Beta(0.5, 0.5)	7.9	34	12.7	3.2	0.8	41.4
U3 - Beta(0.7, 0.3)	8.1	34	12.6	3.4	0.9	41
U3 - Beta(1, 1)	7.5	35	12	3.2	0.8	41.5
Mean administrations						
U2 - Beta(0.3, 0.7)	9	12.1	2.1	0.4	0	-
U2 - Beta(0.5, 0.5)	8.6	9.6	4	1.6	0.4	-
U2 - Beta(0.7, 0.3)	8.5	8.8	4.2	2.2	0.6	-
U2 - Beta(1, 1)	8.5	9.6	4.1	1.6	0.4	-
U3 - Beta(0.3, 0.7)	9.2	12.3	1.9	0.3	0	-
U3 - Beta(0.5, 0.5)	8.7	9.9	3.9	1.4	0.3	-
U3 - Beta(0.7, 0.3)	8.6	9.1	4.2	1.9	0.5	-
U3 - Beta(1, 1)	8.6	9.9	3.9	1.4	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	5	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.8	4.1	1.1	0.6	0.1	-
U2 - Beta(0.7, 0.3)	1.8	3.9	1.1	0.7	0.2	-
U2 - Beta(1, 1)	1.8	4.2	1.1	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.1	5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	1.9	4.2	1.1	0.5	0.1	-
U3 - Beta(0.7, 0.3)	1.9	4	1.1	0.6	0.2	-
U3 - Beta(1, 1)	1.9	4.3	1.1	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						875
U2 - Beta(0.5, 0.5)						902
U2 - Beta(0.7, 0.3)						915.9
U2 - Beta(1, 1)						902.6
U3 - Beta(0.3, 0.7)						874.6
U3 - Beta(0.5, 0.5)						901.6
U3 - Beta(0.7, 0.3)						905.9
U3 - Beta(1, 1)						900.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	58	22.6	4.5	0.2	0	14.7
U2 - Beta(0.5, 0.5)	60.4	20.9	4.2	0.2	0	14.3
U2 - Beta(0.7, 0.3)	59.7	22.6	2.9	0.3	0	14.5
U2 - Beta(1, 1)	59	22.7	4.1	0.2	0	14
U3 - Beta(0.3, 0.7)	61.6	21.2	2	0	0	15.2
U3 - Beta(0.5, 0.5)	62.6	20.4	2.7	0.1	0	14.2
U3 - Beta(0.7, 0.3)	63.3	19.8	2.9	0.1	0	13.9
U3 - Beta(1, 1)	63.1	20	2.4	0.1	0	14.4
Mean administrations						
U2 - Beta(0.3, 0.7)	17.8	7.6	2.2	0.4	0	-
U2 - Beta(0.5, 0.5)	16.8	7.4	2.7	1	0.1	-
U2 - Beta(0.7, 0.3)	16.7	7.6	2.6	1	0.1	-
U2 - Beta(1, 1)	16.5	7.8	2.7	1	0.1	-
U3 - Beta(0.3, 0.7)	18.8	7.2	1.6	0.3	0	-
U3 - Beta(0.5, 0.5)	17.2	7.4	2.6	0.9	0.1	-
U3 - Beta(0.7, 0.3)	17.1	7.5	2.6	1	0.1	-
U3 - Beta(1, 1)	17.3	7.4	2.5	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.4	3.3	0.8	0.2	0	-
U2 - Beta(0.5, 0.5)	4.9	3.4	0.9	0.4	0	-
U2 - Beta(0.7, 0.3)	4.9	3.4	0.8	0.4	0	-
U2 - Beta(1, 1)	4.8	3.5	0.9	0.4	0	-
U3 - Beta(0.3, 0.7)	5.8	3.2	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	5	3.4	0.8	0.3	0	-
U3 - Beta(0.7, 0.3)	5	3.4	0.8	0.4	0	-
U3 - Beta(1, 1)	5.1	3.4	0.7	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008.3
U2 - Beta(0.5, 0.5)						1021.9
U2 - Beta(0.7, 0.3)						1023.6
U2 - Beta(1, 1)						1024.3
U3 - Beta(0.3, 0.7)						1003.3
U3 - Beta(0.5, 0.5)						1021.8
U3 - Beta(0.7, 0.3)						1024.8
U3 - Beta(1, 1)						1023
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	11.4	11.7	58.1	8.4	2.7
U2 - Beta(0.5, 0.5)	11.4	12.4	10.9	46.1	16.7	2.5
U2 - Beta(0.7, 0.3)	13.8	13.1	11.4	42.4	17	2.3
U2 - Beta(1, 1)	12.1	11	10.4	46	18.1	2.4
U3 - Beta(0.3, 0.7)	11.8	13.4	10.5	54.9	6.4	3
U3 - Beta(0.5, 0.5)	15.6	14.8	9.5	43.7	13.9	2.5
U3 - Beta(0.7, 0.3)	15.1	16.2	8.8	41.1	16.4	2.4
U3 - Beta(1, 1)	16.7	14.1	10	41.5	15.2	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	6	6.4	10.8	1.8	-
U2 - Beta(0.5, 0.5)	4.6	5.3	5.9	9.5	4.5	-
U2 - Beta(0.7, 0.3)	4.7	5.4	5.8	8.5	5.3	-
U2 - Beta(1, 1)	4.7	5.4	5.8	9.4	4.5	-
U3 - Beta(0.3, 0.7)	5.1	6.4	6.3	10.4	1.5	-
U3 - Beta(0.5, 0.5)	4.9	5.5	5.8	9.4	4.2	-
U3 - Beta(0.7, 0.3)	5	5.7	5.6	8.6	4.8	-
U3 - Beta(1, 1)	5.1	5.7	5.6	9.2	4.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	3.5	2.3	3.7	0.5	-
U2 - Beta(0.5, 0.5)	0.7	3.3	1.7	3.3	1.2	-
U2 - Beta(0.7, 0.3)	0.7	3.4	1.7	3.1	1.3	-
U2 - Beta(1, 1)	0.7	3.4	1.7	3.3	1.2	-
U3 - Beta(0.3, 0.7)	1	3.6	2.3	3.6	0.4	-
U3 - Beta(0.5, 0.5)	0.8	3.4	1.7	3.3	1.1	-
U3 - Beta(0.7, 0.3)	0.9	3.5	1.6	3	1.3	-
U3 - Beta(1, 1)	0.9	3.5	1.6	3.2	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1043.1
U2 - Beta(0.5, 0.5)						1071.4
U2 - Beta(0.7, 0.3)						1075.3
U2 - Beta(1, 1)						1069.6
U3 - Beta(0.3, 0.7)						1037.1
U3 - Beta(0.5, 0.5)						1067.3
U3 - Beta(0.7, 0.3)						1067.1
U3 - Beta(1, 1)						1065.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.2	34.4	12.9	4	0.1	41.4
U2 - Beta(0.5, 0.5)	9.4	32.5	14.2	4.1	1.2	38.6
U2 - Beta(0.7, 0.3)	9.3	33	13.6	4.9	1.3	37.9
U2 - Beta(1, 1)	9.9	32.2	14	3.6	1.2	39.1
U3 - Beta(0.3, 0.7)	8.4	35.6	10.7	2.9	0.1	42.3
U3 - Beta(0.5, 0.5)	10.5	32.9	12.4	4.1	0.9	39.2
U3 - Beta(0.7, 0.3)	9.7	35	12.5	3.8	1	38
U3 - Beta(1, 1)	11.1	34.1	12.1	3.4	0.8	38.5
Mean administrations						
U2 - Beta(0.3, 0.7)	9.1	11	3.2	0.9	0.1	-
U2 - Beta(0.5, 0.5)	8.5	8.8	4.5	2.2	0.5	-
U2 - Beta(0.7, 0.3)	8.5	8.6	4.4	2.5	0.8	-
U2 - Beta(1, 1)	8.5	8.9	4.4	2.2	0.6	-
U3 - Beta(0.3, 0.7)	9.2	11.2	2.8	0.8	0	-
U3 - Beta(0.5, 0.5)	8.7	9.1	4.3	2.1	0.5	-
U3 - Beta(0.7, 0.3)	8.6	8.9	4.2	2.4	0.6	-
U3 - Beta(1, 1)	8.7	9.2	4.2	2.1	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	4.8	1.1	0.3	0	-
U2 - Beta(0.5, 0.5)	1.9	4.1	1.4	0.8	0.2	-
U2 - Beta(0.7, 0.3)	1.9	4.1	1.3	0.9	0.2	-
U2 - Beta(1, 1)	1.9	4.2	1.3	0.8	0.2	-
U3 - Beta(0.3, 0.7)	2.2	4.9	0.9	0.3	0	-
U3 - Beta(0.5, 0.5)	2	4.2	1.3	0.8	0.2	-
U3 - Beta(0.7, 0.3)	2	4.1	1.3	0.8	0.2	-
U3 - Beta(1, 1)	2	4.2	1.3	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						869.7
U2 - Beta(0.5, 0.5)						893
U2 - Beta(0.7, 0.3)						899.9
U2 - Beta(1, 1)						891.9
U3 - Beta(0.3, 0.7)						863.5
U3 - Beta(0.5, 0.5)						891.4
U3 - Beta(0.7, 0.3)						896.3
U3 - Beta(1, 1)						888.8
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.5	17.9	15.1	39.3	10.1	7.1
U2 - Beta(0.5, 0.5)	15.6	16.7	16.9	31.4	12.6	6.8
U2 - Beta(0.7, 0.3)	17.7	16.8	17.6	30	11.4	6.5
U2 - Beta(1, 1)	17.2	16.4	17.1	30.5	12	6.8
U3 - Beta(0.3, 0.7)	15.7	18.6	13.5	36.4	8.5	7.3
U3 - Beta(0.5, 0.5)	21.5	19.9	15.5	27.7	8.6	6.8
U3 - Beta(0.7, 0.3)	23.9	20.8	14.1	26.1	8.3	6.8
U3 - Beta(1, 1)	24	19.4	15	26.1	8.3	7.2
Mean administrations						
U2 - Beta(0.3, 0.7)	6.1	6.8	6.4	8	2	-
U2 - Beta(0.5, 0.5)	6	6.4	6	7.3	3.7	-
U2 - Beta(0.7, 0.3)	6	6.4	6	7.1	3.9	-
U2 - Beta(1, 1)	6	6.4	5.9	7.3	3.7	-
U3 - Beta(0.3, 0.7)	6.6	7	6.3	7.6	1.8	-
U3 - Beta(0.5, 0.5)	6.4	6.4	5.8	7.3	3.4	-
U3 - Beta(0.7, 0.3)	6.4	6.6	5.7	7.1	3.6	-
U3 - Beta(1, 1)	6.6	6.7	5.7	7.1	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.7	2.3	3	0.7	-
U2 - Beta(0.5, 0.5)	1.3	3.7	1.9	2.8	1.2	-
U2 - Beta(0.7, 0.3)	1.3	3.7	1.9	2.8	1.2	-
U2 - Beta(1, 1)	1.2	3.8	1.9	2.8	1.2	-
U3 - Beta(0.3, 0.7)	1.6	3.8	2.3	2.8	0.7	-
U3 - Beta(0.5, 0.5)	1.5	3.8	1.9	2.8	1.1	-
U3 - Beta(0.7, 0.3)	1.5	3.8	1.8	2.7	1.1	-
U3 - Beta(1, 1)	1.5	3.9	1.8	2.7	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						998.3
U2 - Beta(0.5, 0.5)						1014.6
U2 - Beta(0.7, 0.3)						1018.1
U2 - Beta(1, 1)						1015.8
U3 - Beta(0.3, 0.7)						994.5
U3 - Beta(0.5, 0.5)						1014.1
U3 - Beta(0.7, 0.3)						1013.5
U3 - Beta(1, 1)						1013

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	38.2	30.1	12.8	5.2	0.6	13.1
U2 - Beta(0.5, 0.5)	40	29.4	11.9	5.2	0.4	13.1
U2 - Beta(0.7, 0.3)	40.1	29.8	11.7	4.9	0.6	12.9
U2 - Beta(1, 1)	41.2	27.8	11.7	5.9	0.6	12.8
U3 - Beta(0.3, 0.7)	44.9	31.2	7.4	3.2	0.2	13.1
U3 - Beta(0.5, 0.5)	46.1	28.6	8.2	3.7	0.3	13.1
U3 - Beta(0.7, 0.3)	46.3	28.3	8.5	3.9	0.4	12.6
U3 - Beta(1, 1)	44.4	28.4	9.7	3.9	0.3	13.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.7	8.7	5.9	3.4	0.9	-
U2 - Beta(0.5, 0.5)	9.7	8.4	5.8	3.6	1.1	-
U2 - Beta(0.7, 0.3)	9.9	8.3	5.7	3.6	1.2	-
U2 - Beta(1, 1)	9.7	8.5	5.8	3.6	1.1	-
U3 - Beta(0.3, 0.7)	10.8	8.8	5.2	3.1	0.7	-
U3 - Beta(0.5, 0.5)	10.4	8.3	5.4	3.5	1.1	-
U3 - Beta(0.7, 0.3)	10.4	8.5	5.4	3.4	1.1	-
U3 - Beta(1, 1)	10.4	8.4	5.3	3.4	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.2	2.2	1.4	0.4	-
U2 - Beta(0.5, 0.5)	2.8	4.3	2	1.5	0.4	-
U2 - Beta(0.7, 0.3)	2.8	4.3	1.9	1.5	0.4	-
U2 - Beta(1, 1)	2.8	4.3	2	1.5	0.4	-
U3 - Beta(0.3, 0.7)	3.3	4.3	1.9	1.2	0.3	-
U3 - Beta(0.5, 0.5)	3.1	4.3	1.8	1.5	0.4	-
U3 - Beta(0.7, 0.3)	3.1	4.3	1.8	1.5	0.4	-
U3 - Beta(1, 1)	3.1	4.3	1.7	1.5	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						964.3
U2 - Beta(0.5, 0.5)						971.4
U2 - Beta(0.7, 0.3)						971.5
U2 - Beta(1, 1)						972.4
U3 - Beta(0.3, 0.7)						961.8
U3 - Beta(0.5, 0.5)						970.4
U3 - Beta(0.7, 0.3)						971.7
U3 - Beta(1, 1)						968.6

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.1	22.2	20.5	21.9	14.3	0
U2 - Beta(0.5, 0.5)	38.1	26.6	15.2	12.4	7.7	0
U2 - Beta(0.7, 0.3)	38.4	29.8	15.8	10	6	0
U2 - Beta(1, 1)	40.6	27	15.5	11.1	5.8	0
U3 - Beta(0.3, 0.7)	28.6	27.2	18.8	17.7	7.7	0
U3 - Beta(0.5, 0.5)	43.9	27.9	14	10.4	3.8	0
U3 - Beta(0.7, 0.3)	43.2	30.7	13.7	8.8	3.6	0
U3 - Beta(1, 1)	46.4	29.8	13	7.6	3.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	7.4	5.9	5.3	3	-
U2 - Beta(0.5, 0.5)	4.4	5.1	5.9	8.3	6.2	-
U2 - Beta(0.7, 0.3)	4.3	5	6.1	8.3	6.4	-
U2 - Beta(1, 1)	4.4	5.3	6.3	8.1	6	-
U3 - Beta(0.3, 0.7)	9.1	8	5.8	4.8	2.3	-
U3 - Beta(0.5, 0.5)	4.7	5.3	6.2	8.2	5.6	-
U3 - Beta(0.7, 0.3)	4.8	5.4	6.3	8	5.4	-
U3 - Beta(1, 1)	4.8	5.7	6.4	7.9	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	3.6	2.5	2.2	1.2	-
U2 - Beta(0.5, 0.5)	0.6	2.9	2.3	3.6	2.3	-
U2 - Beta(0.7, 0.3)	0.5	2.9	2.2	3.7	2.3	-
U2 - Beta(1, 1)	0.6	2.9	2.4	3.5	2.3	-
U3 - Beta(0.3, 0.7)	2.7	3.8	2.4	2	1	-
U3 - Beta(0.5, 0.5)	0.7	3	2.3	3.5	2.1	-
U3 - Beta(0.7, 0.3)	0.8	3.1	2.3	3.5	2	-
U3 - Beta(1, 1)	0.7	3.2	2.4	3.4	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						993.4
U2 - Beta(0.5, 0.5)						1008.9
U2 - Beta(0.7, 0.3)						1011.1
U2 - Beta(1, 1)						1007.8
U3 - Beta(0.3, 0.7)						990.1
U3 - Beta(0.5, 0.5)						1006.9
U3 - Beta(0.7, 0.3)						1009.1
U3 - Beta(1, 1)						1006.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.3	53.9	4.9	0.3	0	0.6
U2 - Beta(0.5, 0.5)	20.5	63.5	14.4	1.1	0	0.5
U2 - Beta(0.7, 0.3)	20.7	60.2	15.8	2.7	0	0.6
U2 - Beta(1, 1)	18.4	64.7	15.3	1.1	0	0.5
U3 - Beta(0.3, 0.7)	45.7	52	1.7	0	0	0.6
U3 - Beta(0.5, 0.5)	22.3	65	11.4	0.7	0	0.6
U3 - Beta(0.7, 0.3)	23.1	60.9	13.6	1.9	0	0.5
U3 - Beta(1, 1)	21.2	66.1	11.6	0.7	0	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	17	11.6	1.2	0.1	0	-
U2 - Beta(0.5, 0.5)	9.6	15.3	4.3	0.7	0.1	-
U2 - Beta(0.7, 0.3)	9.2	12.7	5.8	2	0.3	-
U2 - Beta(1, 1)	9.3	15.5	4.4	0.7	0.1	-
U3 - Beta(0.3, 0.7)	17.8	11.5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	10.2	15.5	3.6	0.5	0	-
U3 - Beta(0.7, 0.3)	9.4	13.8	5.1	1.5	0.1	-
U3 - Beta(1, 1)	10.2	15.7	3.6	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.2	3.9	0.3	0	0	-
U2 - Beta(0.5, 0.5)	2.1	5	1.2	0.2	0	-
U2 - Beta(0.7, 0.3)	2	4.3	1.4	0.6	0.1	-
U2 - Beta(1, 1)	2	5.1	1.2	0.2	0	-
U3 - Beta(0.3, 0.7)	5.5	3.9	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2.4	5.2	1	0.2	0	-
U3 - Beta(0.7, 0.3)	2.1	4.6	1.3	0.4	0	-
U3 - Beta(1, 1)	2.3	5.2	1	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1126.2
U2 - Beta(0.5, 0.5)						1178.3
U2 - Beta(0.7, 0.3)						1192.2
U2 - Beta(1, 1)						1179.7
U3 - Beta(0.3, 0.7)						1121
U3 - Beta(0.5, 0.5)						1171.7
U3 - Beta(0.7, 0.3)						1185
U3 - Beta(1, 1)						1172.9

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4	8.2	10.8	26	50.9	0.1
U2 - Beta(0.5, 0.5)	9.9	10.8	8.9	16.4	53.9	0.1
U2 - Beta(0.7, 0.3)	11.8	12.1	11.3	14.6	50.1	0.1
U2 - Beta(1, 1)	13.4	10	10	13.3	53.2	0.1
U3 - Beta(0.3, 0.7)	9.8	10.1	14.3	25.2	40.5	0.1
U3 - Beta(0.5, 0.5)	16.8	14.5	9.8	13.3	45.5	0.1
U3 - Beta(0.7, 0.3)	17.4	15	12.5	12.2	42.8	0.1
U3 - Beta(1, 1)	20.4	15.2	10.2	10.4	43.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.9	6.2	8	7	-
U2 - Beta(0.5, 0.5)	3.5	4	5.2	7.1	10.2	-
U2 - Beta(0.7, 0.3)	3.5	4.1	5.2	6.8	10.4	-
U2 - Beta(1, 1)	3.5	4.2	5.3	6.9	10.2	-
U3 - Beta(0.3, 0.7)	3.9	5.1	6.7	8.1	6.1	-
U3 - Beta(0.5, 0.5)	3.5	4.2	5.3	7.2	9.7	-
U3 - Beta(0.7, 0.3)	3.8	4.3	5.3	6.8	9.7	-
U3 - Beta(1, 1)	3.8	4.4	5.3	7	9.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3	2.5	3.2	2.2	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	2.9	2.9	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.7	2.9	2.9	-
U2 - Beta(1, 1)	0.2	2.9	1.8	2.9	2.9	-
U3 - Beta(0.3, 0.7)	0.5	3.1	2.7	3.2	1.9	-
U3 - Beta(0.5, 0.5)	0.2	2.9	1.8	3	2.8	-
U3 - Beta(0.7, 0.3)	0.4	3	1.8	2.8	2.8	-
U3 - Beta(1, 1)	0.4	3	1.8	2.9	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1027.2
U2 - Beta(0.5, 0.5)						1061.8
U2 - Beta(0.7, 0.3)						1064.3
U2 - Beta(1, 1)						1060.6
U3 - Beta(0.3, 0.7)						1018.3
U3 - Beta(0.5, 0.5)						1058.1
U3 - Beta(0.7, 0.3)						1057.7
U3 - Beta(1, 1)						1054.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.4	57.6	17	0.7	0	13.3
U2 - Beta(0.5, 0.5)	11.8	57.2	15.8	0.7	0	14.5
U2 - Beta(0.7, 0.3)	12.4	55.2	16.3	1.6	0	14.5
U2 - Beta(1, 1)	12.1	56.5	16.2	0.9	0	14.3
U3 - Beta(0.3, 0.7)	12.6	60.8	12.8	0.7	0	13.1
U3 - Beta(0.5, 0.5)	13.7	56.9	14.1	0.6	0	14.7
U3 - Beta(0.7, 0.3)	13	55.9	14.7	1.8	0	14.6
U3 - Beta(1, 1)	14.6	56.2	14.5	1	0	13.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	15.2	4.6	0.6	0	-
U2 - Beta(0.5, 0.5)	7.9	12.6	5.9	1.9	0.2	-
U2 - Beta(0.7, 0.3)	7.9	12.2	5.8	2.2	0.3	-
U2 - Beta(1, 1)	7.8	12.7	5.9	1.8	0.2	-
U3 - Beta(0.3, 0.7)	8.5	15.6	3.9	0.5	0	-
U3 - Beta(0.5, 0.5)	8.1	12.8	5.7	1.7	0.2	-
U3 - Beta(0.7, 0.3)	8	12.5	5.7	2	0.2	-
U3 - Beta(1, 1)	8.2	12.9	5.6	1.7	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	6.2	1.5	0.2	0	-
U2 - Beta(0.5, 0.5)	1.7	5.5	1.7	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.7	5.4	1.6	0.8	0.1	-
U2 - Beta(1, 1)	1.6	5.5	1.7	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.1	6.4	1.3	0.2	0	-
U3 - Beta(0.5, 0.5)	1.8	5.6	1.7	0.6	0.1	-
U3 - Beta(0.7, 0.3)	1.8	5.5	1.6	0.7	0.1	-
U3 - Beta(1, 1)	1.8	5.7	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1019.8
U2 - Beta(0.5, 0.5)						1032.7
U2 - Beta(0.7, 0.3)						1038.3
U2 - Beta(1, 1)						1033.2
U3 - Beta(0.3, 0.7)						1016.2
U3 - Beta(0.5, 0.5)						1029.1
U3 - Beta(0.7, 0.3)						1036.2
U3 - Beta(1, 1)						1029.2
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	49.6	30.7	11.5	1	2.6
U2 - Beta(0.5, 0.5)	8.4	48.8	30.1	8.9	1.3	2.5
U2 - Beta(0.7, 0.3)	9.4	48	27.7	11.3	1	2.6
U2 - Beta(1, 1)	9.9	47.5	28.4	10.7	1.2	2.3
U3 - Beta(0.3, 0.7)	7.2	55.5	26.3	8.5	0.2	2.3
U3 - Beta(0.5, 0.5)	10.1	53.8	24.6	8.2	0.8	2.5
U3 - Beta(0.7, 0.3)	11	52.4	24.3	8.7	0.7	2.9
U3 - Beta(1, 1)	14	52	21.4	8.9	1	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5	13.3	7.8	3.1	0.5	-
U2 - Beta(0.5, 0.5)	4.8	10.5	8.3	4.6	1.4	-
U2 - Beta(0.7, 0.3)	5	10.2	8	4.9	1.6	-
U2 - Beta(1, 1)	5	10.6	8.1	4.6	1.4	-
U3 - Beta(0.3, 0.7)	5.3	14.4	7.2	2.5	0.3	-
U3 - Beta(0.5, 0.5)	5.1	11.3	7.7	4.3	1.3	-
U3 - Beta(0.7, 0.3)	5.3	10.8	7.7	4.5	1.3	-
U3 - Beta(1, 1)	5.3	11.4	7.4	4.3	1.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	5.9	2.8	1.2	0.2	-
U2 - Beta(0.5, 0.5)	0.7	5	2.6	1.7	0.5	-
U2 - Beta(0.7, 0.3)	0.8	5	2.4	1.8	0.5	-
U2 - Beta(1, 1)	0.8	5	2.6	1.7	0.5	-
U3 - Beta(0.3, 0.7)	1	6.4	2.5	1	0.1	-
U3 - Beta(0.5, 0.5)	0.9	5.3	2.4	1.6	0.4	-
U3 - Beta(0.7, 0.3)	0.9	5.2	2.4	1.6	0.5	-
U3 - Beta(1, 1)	0.9	5.4	2.3	1.6	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.7
U2 - Beta(0.5, 0.5)						1052.2
U2 - Beta(0.7, 0.3)						1052.9
U2 - Beta(1, 1)						1051
U3 - Beta(0.3, 0.7)						1029.5
U3 - Beta(0.5, 0.5)						1046.8
U3 - Beta(0.7, 0.3)						1050.1
U3 - Beta(1, 1)						1048

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	56.4	0.2	0	0	0	43.4
U2 - Beta(0.5, 0.5)	52	1.1	0	0	0	46.9
U2 - Beta(0.7, 0.3)	50.8	1.3	0	0	0	47.9
U2 - Beta(1, 1)	52.4	1	0	0	0	46.6
U3 - Beta(0.3, 0.7)	56.1	0.1	0	0	0	43.8
U3 - Beta(0.5, 0.5)	52.6	1	0	0	0	46.4
U3 - Beta(0.7, 0.3)	52.4	1.3	0	0	0	46.3
U3 - Beta(1, 1)	52.2	1	0	0	0	46.8
Mean administrations						
U2 - Beta(0.3, 0.7)	22.2	0.5	0	0	0	-
U2 - Beta(0.5, 0.5)	19.8	2.3	0.1	0	0	-
U2 - Beta(0.7, 0.3)	18.6	3.4	0.2	0	0	-
U2 - Beta(1, 1)	19.9	2.3	0.1	0	0	-
U3 - Beta(0.3, 0.7)	22.2	0.5	0	0	0	-
U3 - Beta(0.5, 0.5)	19.9	2.3	0.1	0	0	-
U3 - Beta(0.7, 0.3)	19.2	2.8	0.2	0	0	-
U3 - Beta(1, 1)	19.9	2.2	0.1	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.7	0.3	0	0	0	-
U2 - Beta(0.5, 0.5)	3.8	0.9	0	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1.2	0.1	0	0	-
U2 - Beta(1, 1)	3.8	0.9	0	0	0	-
U3 - Beta(0.3, 0.7)	4.7	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.8	0.9	0	0	0	-
U3 - Beta(0.7, 0.3)	3.6	1.1	0.1	0	0	-
U3 - Beta(1, 1)	3.8	0.9	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						982.7
U2 - Beta(0.5, 0.5)						966.7
U2 - Beta(0.7, 0.3)						967.1
U2 - Beta(1, 1)						968.8
U3 - Beta(0.3, 0.7)						982.4
U3 - Beta(0.5, 0.5)						967.9
U3 - Beta(0.7, 0.3)						971.5
U3 - Beta(1, 1)						965.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.2	62.2	12.2	1.2	0.1	11.1
U2 - Beta(0.5, 0.5)	14.6	60.8	12	1.4	0	11.2
U2 - Beta(0.7, 0.3)	15	59.5	11.9	2	0	11.6
U2 - Beta(1, 1)	14.8	60.2	12.4	1.5	0	11.1
U3 - Beta(0.3, 0.7)	16.3	62.2	9.7	0.8	0.1	10.9
U3 - Beta(0.5, 0.5)	16.1	60.7	9.7	1.5	0	12
U3 - Beta(0.7, 0.3)	16.9	59.1	10.1	2.2	0	11.7
U3 - Beta(1, 1)	17.7	59.2	10.3	1.9	0	10.9
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	15.3	4.5	0.9	0	-
U2 - Beta(0.5, 0.5)	8.4	13.3	5.4	1.8	0.1	-
U2 - Beta(0.7, 0.3)	8.4	13	5.3	2.1	0.2	-
U2 - Beta(1, 1)	8.3	13.4	5.4	1.9	0.1	-
U3 - Beta(0.3, 0.7)	8.9	15.5	3.9	0.7	0	-
U3 - Beta(0.5, 0.5)	8.7	13.4	5	1.7	0.1	-
U3 - Beta(0.7, 0.3)	8.6	13.1	5.1	2	0.1	-
U3 - Beta(1, 1)	8.7	13.4	5.1	1.7	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.4	1.6	0.3	0	-
U2 - Beta(0.5, 0.5)	2	5.9	1.6	0.6	0	-
U2 - Beta(0.7, 0.3)	2	5.8	1.5	0.7	0.1	-
U2 - Beta(1, 1)	1.9	5.9	1.6	0.7	0	-
U3 - Beta(0.3, 0.7)	2.3	6.5	1.4	0.2	0	-
U3 - Beta(0.5, 0.5)	2.1	5.9	1.5	0.6	0	-
U3 - Beta(0.7, 0.3)	2.1	5.8	1.4	0.7	0	-
U3 - Beta(1, 1)	2.1	5.9	1.4	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1026.6
U2 - Beta(0.5, 0.5)						1039.7
U2 - Beta(0.7, 0.3)						1040.4
U2 - Beta(1, 1)						1038.5
U3 - Beta(0.3, 0.7)						1019.1
U3 - Beta(0.5, 0.5)						1033.6
U3 - Beta(0.7, 0.3)						1039
U3 - Beta(1, 1)						1034.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.4	0.8	0	0	0	91.8
U2 - Beta(0.5, 0.5)	7.4	0.6	0	0	0	92
U2 - Beta(0.7, 0.3)	7.4	1.2	0	0	0	91.4
U2 - Beta(1, 1)	7.3	1.1	0	0	0	91.6
U3 - Beta(0.3, 0.7)	6.9	0.8	0	0	0	92.3
U3 - Beta(0.5, 0.5)	7.2	0.8	0	0	0	92
U3 - Beta(0.7, 0.3)	7	1	0	0	0	92
U3 - Beta(1, 1)	7.8	0.8	0	0	0	91.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	3.7	0.5	0	0	-
U2 - Beta(0.5, 0.5)	11.1	3.9	0.5	0	0	-
U2 - Beta(0.7, 0.3)	11.1	3.9	0.5	0.1	0	-
U2 - Beta(1, 1)	11	3.9	0.5	0	0	-
U3 - Beta(0.3, 0.7)	11.3	3.6	0.4	0	0	-
U3 - Beta(0.5, 0.5)	11.1	3.8	0.5	0.1	0	-
U3 - Beta(0.7, 0.3)	11.1	3.9	0.5	0.1	0	-
U3 - Beta(1, 1)	11	3.9	0.5	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2.1	0.2	0	0	-
U2 - Beta(0.7, 0.3)	2.9	2.1	0.2	0	0	-
U2 - Beta(1, 1)	2.8	2.1	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3	2	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	2.1	0.2	0	0	-
U3 - Beta(0.7, 0.3)	2.9	2.1	0.2	0	0	-
U3 - Beta(1, 1)	2.9	2.1	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						562.5
U2 - Beta(0.5, 0.5)						559
U2 - Beta(0.7, 0.3)						560.4
U2 - Beta(1, 1)						558.3
U3 - Beta(0.3, 0.7)						558.2
U3 - Beta(0.5, 0.5)						558.3
U3 - Beta(0.7, 0.3)						558.5
U3 - Beta(1, 1)						559.6
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	0.7	8.6	75.4	15.1	0
U2 - Beta(0.5, 0.5)	1.5	0.9	2.5	59.2	35.9	0
U2 - Beta(0.7, 0.3)	3.3	2.3	5.5	48.9	40	0
U2 - Beta(1, 1)	1.8	1.4	3.3	57.5	36	0
U3 - Beta(0.3, 0.7)	0.8	1.2	11.1	76.8	10.1	0
U3 - Beta(0.5, 0.5)	2.6	2.1	4	62	29.3	0
U3 - Beta(0.7, 0.3)	3.9	3.7	5.8	51.3	35.3	0
U3 - Beta(1, 1)	3.2	2.7	4	61.8	28.3	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4.1	6.8	13.5	1.9	-
U2 - Beta(0.5, 0.5)	3.1	3.2	3.8	13	7	-
U2 - Beta(0.7, 0.3)	3.1	3.1	3.8	11	9	-
U2 - Beta(1, 1)	3.1	3.2	3.8	12.9	7	-
U3 - Beta(0.3, 0.7)	3.6	4.2	7.1	13.6	1.4	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	13.6	6.1	-
U3 - Beta(0.7, 0.3)	3.1	3.2	4	11.6	8.1	-
U3 - Beta(1, 1)	3.1	3.2	4	13.6	6.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	2.7	4.6	0.6	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.5	2.1	-
U2 - Beta(0.7, 0.3)	0	2.4	1.2	4	2.5	-
U2 - Beta(1, 1)	0.1	2.4	1.2	4.4	2.1	-
U3 - Beta(0.3, 0.7)	0.4	2.7	2.9	4.7	0.5	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.2	4.7	1.8	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.2	4.1	2.3	-
U3 - Beta(1, 1)	0.1	2.5	1.3	4.7	1.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1046.4
U2 - Beta(0.5, 0.5)						1090.6
U2 - Beta(0.7, 0.3)						1096.3
U2 - Beta(1, 1)						1091.1
U3 - Beta(0.3, 0.7)						1043.4
U3 - Beta(0.5, 0.5)						1087.5
U3 - Beta(0.7, 0.3)						1090.8
U3 - Beta(1, 1)						1087.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	24.8	38.7	33	2.2	0.1	1.2
U2 - Beta(0.5, 0.5)	28.6	33	33.6	3.3	0.1	1.4
U2 - Beta(0.7, 0.3)	28.5	33.4	31.9	4.6	0	1.6
U2 - Beta(1, 1)	28.4	32.9	33.9	3.3	0.1	1.4
U3 - Beta(0.3, 0.7)	30.7	42.5	23.9	1.6	0	1.3
U3 - Beta(0.5, 0.5)	33	33.7	29.1	2.9	0	1.3
U3 - Beta(0.7, 0.3)	33.3	33.3	27.5	4.5	0	1.4
U3 - Beta(1, 1)	33.6	33	28.7	3.1	0.1	1.5
Mean administrations						
U2 - Beta(0.3, 0.7)	10.9	11.1	7	0.8	0	-
U2 - Beta(0.5, 0.5)	9	9.8	8.5	2.4	0.2	-
U2 - Beta(0.7, 0.3)	9.2	9.7	7.7	2.9	0.4	-
U2 - Beta(1, 1)	8.8	9.8	8.6	2.4	0.2	-
U3 - Beta(0.3, 0.7)	12.2	11.8	5.5	0.5	0	-
U3 - Beta(0.5, 0.5)	9.4	10	8.1	2.1	0.2	-
U3 - Beta(0.7, 0.3)	9.4	10	7.6	2.6	0.3	-
U3 - Beta(1, 1)	9.7	10	7.9	2.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	4.6	2.2	0.3	0	-
U2 - Beta(0.5, 0.5)	2.3	4.2	2.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.4	4.3	2	0.9	0.1	-
U2 - Beta(1, 1)	2.2	4.2	2.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	3.8	4.9	1.7	0.2	0	-
U3 - Beta(0.5, 0.5)	2.5	4.3	2.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	4.3	2	0.9	0.1	-
U3 - Beta(1, 1)	2.6	4.4	2.2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1073.2
U2 - Beta(0.5, 0.5)						1107.9
U2 - Beta(0.7, 0.3)						1113.2
U2 - Beta(1, 1)						1109.5
U3 - Beta(0.3, 0.7)						1061.7
U3 - Beta(0.5, 0.5)						1104.5
U3 - Beta(0.7, 0.3)						1107.6
U3 - Beta(1, 1)						1102.2

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17	32.9	36.3	2.1	0	11.7
U2 - Beta(0.5, 0.5)	19.8	33.5	28.9	5.1	0.2	12.5
U2 - Beta(0.7, 0.3)	19.8	35.1	28.7	4	0.1	12.3
U2 - Beta(1, 1)	19.2	33.4	30.1	5.2	0.2	11.9
U3 - Beta(0.3, 0.7)	20.7	33.8	31.3	2	0	12.2
U3 - Beta(0.5, 0.5)	23.1	35.2	25.6	4	0.3	11.8
U3 - Beta(0.7, 0.3)	24.9	33	24.3	4.8	0.2	12.8
U3 - Beta(1, 1)	25.2	34.1	23.7	4	0.2	12.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.3	7.9	1.2	0	-
U2 - Beta(0.5, 0.5)	8.6	9.2	7.7	2.6	0.3	-
U2 - Beta(0.7, 0.3)	8.9	9.2	7.1	2.9	0.4	-
U2 - Beta(1, 1)	8.4	9.3	7.9	2.7	0.3	-
U3 - Beta(0.3, 0.7)	9.8	10.6	7.1	0.9	0	-
U3 - Beta(0.5, 0.5)	9	9.3	7.5	2.4	0.2	-
U3 - Beta(0.7, 0.3)	9.1	9.3	7	2.8	0.2	-
U3 - Beta(1, 1)	9.1	9.5	7.3	2.4	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	4.6	2.6	0.5	0	-
U2 - Beta(0.5, 0.5)	2.1	4.4	2.2	1	0.1	-
U2 - Beta(0.7, 0.3)	2.2	4.4	1.9	1	0.1	-
U2 - Beta(1, 1)	2	4.4	2.2	1	0.1	-
U3 - Beta(0.3, 0.7)	2.7	4.7	2.3	0.4	0	-
U3 - Beta(0.5, 0.5)	2.3	4.4	2.1	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.3	4.4	2	1	0.1	-
U3 - Beta(1, 1)	2.3	4.5	2	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1005
U2 - Beta(0.5, 0.5)						1025.8
U2 - Beta(0.7, 0.3)						1031.4
U2 - Beta(1, 1)						1029.9
U3 - Beta(0.3, 0.7)						995.9
U3 - Beta(0.5, 0.5)						1024.9
U3 - Beta(0.7, 0.3)						1027.4
U3 - Beta(1, 1)						1021.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	27.3	50.8	6	0.3	4.9
U2 - Beta(0.5, 0.5)	14.3	24.5	46.2	10	0.3	4.7
U2 - Beta(0.7, 0.3)	15.3	25.5	43.6	9.8	0.5	5.3
U2 - Beta(1, 1)	14.4	26	44.7	10.3	0.3	4.3
U3 - Beta(0.3, 0.7)	16	30.2	43.9	4.4	0.3	5.2
U3 - Beta(0.5, 0.5)	17.8	27.6	42.1	7.6	0.3	4.6
U3 - Beta(0.7, 0.3)	18.6	26.5	41.7	8.2	0.2	4.8
U3 - Beta(1, 1)	18.8	27.3	41.4	7.6	0.2	4.7
Mean administrations						
U2 - Beta(0.3, 0.7)	7.2	10.2	10.3	1.6	0.1	-
U2 - Beta(0.5, 0.5)	6.7	8.6	10	3.5	0.6	-
U2 - Beta(0.7, 0.3)	6.8	8.5	9.1	4	0.9	-
U2 - Beta(1, 1)	6.5	8.7	10	3.6	0.6	-
U3 - Beta(0.3, 0.7)	8.1	10.7	9.3	1.2	0.1	-
U3 - Beta(0.5, 0.5)	7	8.8	9.9	3.2	0.5	-
U3 - Beta(0.7, 0.3)	7	8.8	9.1	3.9	0.6	-
U3 - Beta(1, 1)	7.1	8.9	9.7	3.2	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	4.6	3.4	0.6	0	-
U2 - Beta(0.5, 0.5)	1.5	4.1	2.9	1.2	0.2	-
U2 - Beta(0.7, 0.3)	1.5	4.1	2.6	1.3	0.3	-
U2 - Beta(1, 1)	1.4	4.1	2.9	1.2	0.2	-
U3 - Beta(0.3, 0.7)	2.2	4.8	3	0.4	0	-
U3 - Beta(0.5, 0.5)	1.6	4.2	2.9	1.1	0.2	-
U3 - Beta(0.7, 0.3)	1.6	4.2	2.6	1.3	0.2	-
U3 - Beta(1, 1)	1.6	4.2	2.8	1.1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1047.4
U2 - Beta(0.5, 0.5)						1077.2
U2 - Beta(0.7, 0.3)						1082
U2 - Beta(1, 1)						1077.6
U3 - Beta(0.3, 0.7)						1035.6
U3 - Beta(0.5, 0.5)						1072
U3 - Beta(0.7, 0.3)						1077.9
U3 - Beta(1, 1)						1072.9
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.5	41.3	4.5	0.2	0	44.5
U2 - Beta(0.5, 0.5)	10.6	38.9	6.1	0.1	0	44.3
U2 - Beta(0.7, 0.3)	10.9	38.6	6	0	0	44.5
U2 - Beta(1, 1)	11.6	38	5.7	0.1	0	44.6
U3 - Beta(0.3, 0.7)	9.1	42.1	3.7	0.1	0	45
U3 - Beta(0.5, 0.5)	11.9	39.2	4.4	0	0	44.5
U3 - Beta(0.7, 0.3)	11.4	39.1	5.3	0.1	0	44.1
U3 - Beta(1, 1)	12	38	5.5	0	0	44.5
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	12.2	2.2	0.3	0	-
U2 - Beta(0.5, 0.5)	9.4	10.6	3.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	9.5	10.3	3.4	1.1	0.1	-
U2 - Beta(1, 1)	9.4	10.7	3.3	0.8	0.1	-
U3 - Beta(0.3, 0.7)	9.6	12.3	1.9	0.2	0	-
U3 - Beta(0.5, 0.5)	9.7	10.6	3.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	9.6	10.5	3.2	1	0.1	-
U3 - Beta(1, 1)	9.7	10.7	3.1	0.8	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	5.2	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	2.2	4.6	1	0.3	0	-
U2 - Beta(0.7, 0.3)	2.3	4.6	1	0.4	0	-
U2 - Beta(1, 1)	2.2	4.7	1	0.3	0	-
U3 - Beta(0.3, 0.7)	2.4	5.2	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	2.4	4.6	1	0.3	0	-
U3 - Beta(0.7, 0.3)	2.3	4.6	0.9	0.3	0	-
U3 - Beta(1, 1)	2.4	4.7	0.9	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						870.6
U2 - Beta(0.5, 0.5)						881.3
U2 - Beta(0.7, 0.3)						886.2
U2 - Beta(1, 1)						883.7
U3 - Beta(0.3, 0.7)						863.5
U3 - Beta(0.5, 0.5)						878.5
U3 - Beta(0.7, 0.3)						887.2
U3 - Beta(1, 1)						880.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	80.7	16.3	0.7	0	0	2.3
U2 - Beta(0.5, 0.5)	64.5	30.1	2.9	0.1	0	2.4
U2 - Beta(0.7, 0.3)	66	28.5	3.1	0.2	0	2.2
U2 - Beta(1, 1)	63.9	30.7	3	0.1	0	2.3
U3 - Beta(0.3, 0.7)	82.5	14.9	0.2	0	0	2.4
U3 - Beta(0.5, 0.5)	68.3	27.5	1.7	0.1	0	2.4
U3 - Beta(0.7, 0.3)	67.3	27.4	2.9	0.2	0	2.2
U3 - Beta(1, 1)	67	28.9	1.7	0.1	0	2.3
Mean administrations						
U2 - Beta(0.3, 0.7)	24.9	4.2	0.4	0	0	-
U2 - Beta(0.5, 0.5)	18.1	8.9	2.1	0.4	0	-
U2 - Beta(0.7, 0.3)	17.5	8.9	2.5	0.7	0	-
U2 - Beta(1, 1)	18	9.1	2	0.4	0	-
U3 - Beta(0.3, 0.7)	25.2	4	0.3	0	0	-
U3 - Beta(0.5, 0.5)	19	8.6	1.7	0.3	0	-
U3 - Beta(0.7, 0.3)	17.7	8.9	2.3	0.6	0	-
U3 - Beta(1, 1)	19.2	8.5	1.5	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7	1.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.4	3.2	0.6	0.1	0	-
U2 - Beta(0.7, 0.3)	4.3	3.1	0.6	0.2	0	-
U2 - Beta(1, 1)	4.4	3.2	0.6	0.1	0	-
U3 - Beta(0.3, 0.7)	7.1	1.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.8	3.1	0.5	0.1	0	-
U3 - Beta(0.7, 0.3)	4.3	3.2	0.7	0.2	0	-
U3 - Beta(1, 1)	4.8	3.1	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1149.7
U2 - Beta(0.5, 0.5)						1167.6
U2 - Beta(0.7, 0.3)						1179.5
U2 - Beta(1, 1)						1169.5
U3 - Beta(0.3, 0.7)						1147.9
U3 - Beta(0.5, 0.5)						1164.8
U3 - Beta(0.7, 0.3)						1174.1
U3 - Beta(1, 1)						1164.4
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.1	11	18.9	55.5	5.1	0.4
U2 - Beta(0.5, 0.5)	15.7	11.7	15.4	45.6	11.2	0.4
U2 - Beta(0.7, 0.3)	18.9	16.1	13	40.2	11.4	0.4
U2 - Beta(1, 1)	18.4	12.2	14.2	43.1	11.7	0.4
U3 - Beta(0.3, 0.7)	16.6	14.4	19.1	46.4	3.1	0.4
U3 - Beta(0.5, 0.5)	22.8	16.3	12.4	39	9	0.5
U3 - Beta(0.7, 0.3)	23.9	18.6	13.3	34.3	9.5	0.4
U3 - Beta(1, 1)	26.4	16.5	12.9	36.3	7.5	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	6.7	7.5	8.9	1.2	-
U2 - Beta(0.5, 0.5)	4.4	5.2	6.4	10.2	3.7	-
U2 - Beta(0.7, 0.3)	4.7	5.1	6.4	9.4	4.4	-
U2 - Beta(1, 1)	4.4	5.3	6.4	10.1	3.8	-
U3 - Beta(0.3, 0.7)	6.6	7.3	7.2	8	0.8	-
U3 - Beta(0.5, 0.5)	4.7	5.4	6.4	10	3.4	-
U3 - Beta(0.7, 0.3)	4.9	5.5	6.3	9.3	3.9	-
U3 - Beta(1, 1)	4.9	5.5	6.3	10	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	3.5	2.9	3	0.4	-
U2 - Beta(0.5, 0.5)	0.6	3.1	2	3.4	1.1	-
U2 - Beta(0.7, 0.3)	0.7	3.1	1.9	3.2	1.2	-
U2 - Beta(1, 1)	0.6	3.1	2	3.3	1.2	-
U3 - Beta(0.3, 0.7)	1.7	3.7	2.8	2.7	0.3	-
U3 - Beta(0.5, 0.5)	0.7	3.2	2	3.3	1.1	-
U3 - Beta(0.7, 0.3)	0.8	3.3	2	3.1	1.2	-
U3 - Beta(1, 1)	0.8	3.2	2	3.3	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1040.9
U2 - Beta(0.5, 0.5)						1087.1
U2 - Beta(0.7, 0.3)						1089.6
U2 - Beta(1, 1)						1084.5
U3 - Beta(0.3, 0.7)						1030.3
U3 - Beta(0.5, 0.5)						1082.3
U3 - Beta(0.7, 0.3)						1082.3
U3 - Beta(1, 1)						1079

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.8	74.9	16.1	2	0.2	0
U2 - Beta(0.5, 0.5)	2.7	39.2	30.4	23.3	4.4	0
U2 - Beta(0.7, 0.3)	5.9	34.2	25.1	27.3	7.5	0
U2 - Beta(1, 1)	3.6	37.8	30.8	23.4	4.4	0
U3 - Beta(0.3, 0.7)	9.3	76.9	12.5	1.2	0.1	0
U3 - Beta(0.5, 0.5)	4.9	42.6	29.4	20.6	2.5	0
U3 - Beta(0.7, 0.3)	7.5	38.3	24.3	25.5	4.4	0
U3 - Beta(1, 1)	5.7	43.9	28.9	18.9	2.6	0
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	18.3	3.1	0.3	0	-
U2 - Beta(0.5, 0.5)	3.8	12.6	7.7	4.7	1.2	-
U2 - Beta(0.7, 0.3)	3.6	6.9	7.6	8.3	3.6	-
U2 - Beta(1, 1)	3.8	12.4	7.8	4.7	1.2	-
U3 - Beta(0.3, 0.7)	8.6	18.6	2.5	0.2	0	-
U3 - Beta(0.5, 0.5)	4	13.3	7.4	4.5	0.8	-
U3 - Beta(0.7, 0.3)	3.8	8.8	7.8	7.1	2.5	-
U3 - Beta(1, 1)	4	13.5	7.4	4.3	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	6.6	1	0.1	0	-
U2 - Beta(0.5, 0.5)	0.3	4.9	2.3	1.4	0.4	-
U2 - Beta(0.7, 0.3)	0.2	3.2	2.1	2.3	1	-
U2 - Beta(1, 1)	0.3	4.9	2.3	1.4	0.4	-
U3 - Beta(0.3, 0.7)	2.5	6.7	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	0.3	5.2	2.2	1.3	0.2	-
U3 - Beta(0.7, 0.3)	0.3	3.7	2.3	2	0.7	-
U3 - Beta(1, 1)	0.3	5.2	2.2	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1103.9
U2 - Beta(0.5, 0.5)						1144.5
U2 - Beta(0.7, 0.3)						1171.4
U2 - Beta(1, 1)						1145.4
U3 - Beta(0.3, 0.7)						1101.9
U3 - Beta(0.5, 0.5)						1140.8
U3 - Beta(0.7, 0.3)						1160
U3 - Beta(1, 1)						1140.2
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.1	8.9	45	33.8	9.1	0.1
U2 - Beta(0.5, 0.5)	11.9	13.4	34.9	29.2	10.4	0.2
U2 - Beta(0.7, 0.3)	13.6	17.2	32.2	26.8	10.1	0.1
U2 - Beta(1, 1)	14.4	15.6	30.8	29.1	10	0.1
U3 - Beta(0.3, 0.7)	6.7	12.3	45.3	29.8	5.8	0.1
U3 - Beta(0.5, 0.5)	16.7	18	32.7	27.4	5.1	0.1
U3 - Beta(0.7, 0.3)	17.7	18.9	32.5	24.9	5.9	0.1
U3 - Beta(1, 1)	20.4	19.1	30.1	25.4	4.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	6	11.7	6.4	1.7	-
U2 - Beta(0.5, 0.5)	3.5	4.3	8.7	8.7	4.7	-
U2 - Beta(0.7, 0.3)	3.6	4.4	8.6	8.4	5	-
U2 - Beta(1, 1)	3.5	4.4	8.7	8.7	4.6	-
U3 - Beta(0.3, 0.7)	4.4	6.6	11.7	6	1.3	-
U3 - Beta(0.5, 0.5)	3.6	4.7	9	8.6	4.1	-
U3 - Beta(0.7, 0.3)	3.7	4.9	8.9	8.2	4.3	-
U3 - Beta(1, 1)	3.8	4.9	9	8.5	3.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.3	4.4	2.4	0.6	-
U2 - Beta(0.5, 0.5)	0.2	2.9	2.9	3.3	1.5	-
U2 - Beta(0.7, 0.3)	0.2	2.9	2.8	3.2	1.6	-
U2 - Beta(1, 1)	0.2	2.9	2.9	3.3	1.5	-
U3 - Beta(0.3, 0.7)	0.7	3.6	4.4	2.2	0.5	-
U3 - Beta(0.5, 0.5)	0.3	3	3	3.2	1.3	-
U3 - Beta(0.7, 0.3)	0.3	3.1	2.9	3.1	1.4	-
U3 - Beta(1, 1)	0.4	3.1	3	3.2	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.1
U2 - Beta(0.5, 0.5)						1055.2
U2 - Beta(0.7, 0.3)						1058.7
U2 - Beta(1, 1)						1055
U3 - Beta(0.3, 0.7)						1023.6
U3 - Beta(0.5, 0.5)						1051.7
U3 - Beta(0.7, 0.3)						1053.4
U3 - Beta(1, 1)						1050.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3	4.1	17.1	60.9	14.8	0.1
U2 - Beta(0.5, 0.5)	8.4	5.8	11.4	47.7	26.6	0.1
U2 - Beta(0.7, 0.3)	9.4	7.2	13.1	43.6	26.6	0.1
U2 - Beta(1, 1)	10.1	6.6	10.7	47.3	25.2	0.1
U3 - Beta(0.3, 0.7)	6.6	4.3	18.5	60.7	9.8	0.1
U3 - Beta(0.5, 0.5)	12.6	8.4	11.7	45.7	21.5	0.1
U3 - Beta(0.7, 0.3)	14.7	8.2	13	41.2	22.8	0.1
U3 - Beta(1, 1)	13.8	8.7	11.4	45.2	20.8	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.4	7.9	11.5	2.4	-
U2 - Beta(0.5, 0.5)	3.3	3.8	6.1	10.7	6.2	-
U2 - Beta(0.7, 0.3)	3.4	3.7	5.8	10	7.1	-
U2 - Beta(1, 1)	3.4	3.8	6	10.7	6.1	-
U3 - Beta(0.3, 0.7)	3.8	4.5	8.3	11.5	1.8	-
U3 - Beta(0.5, 0.5)	3.5	3.9	6.1	10.9	5.6	-
U3 - Beta(0.7, 0.3)	3.7	4	6	10.1	6.3	-
U3 - Beta(1, 1)	3.7	4	6	10.8	5.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	3	3.9	0.7	-
U2 - Beta(0.5, 0.5)	0.1	2.7	1.8	3.8	1.7	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.7	1.8	-
U2 - Beta(1, 1)	0.2	2.7	1.8	3.8	1.7	-
U3 - Beta(0.3, 0.7)	0.5	2.9	3.1	3.9	0.6	-
U3 - Beta(0.5, 0.5)	0.2	2.8	1.8	3.9	1.5	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.8	3.6	1.6	-
U3 - Beta(1, 1)	0.3	2.9	1.8	3.8	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1056.2
U2 - Beta(0.5, 0.5)						1089.2
U2 - Beta(0.7, 0.3)						1093.9
U2 - Beta(1, 1)						1087.8
U3 - Beta(0.3, 0.7)						1050.9
U3 - Beta(0.5, 0.5)						1086.8
U3 - Beta(0.7, 0.3)						1086.5
U3 - Beta(1, 1)						1083.9
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	12.5	54.2	17.9	2.8	4.9
U2 - Beta(0.5, 0.5)	13.2	12.9	47.7	19	3.2	4
U2 - Beta(0.7, 0.3)	13.7	14	44.5	19.5	4	4.3
U2 - Beta(1, 1)	14.2	13	46.8	19.4	2.7	3.9
U3 - Beta(0.3, 0.7)	10.3	14.1	52.2	16.7	1.7	5
U3 - Beta(0.5, 0.5)	16.2	15.9	44.2	17.6	2	4.1
U3 - Beta(0.7, 0.3)	18.9	16.1	39.9	18.6	2.4	4.1
U3 - Beta(1, 1)	20.1	15.3	41.1	17.4	2.2	3.9
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	6.9	12.6	4.3	0.6	-
U2 - Beta(0.5, 0.5)	5.1	6.6	9.9	5.9	2.1	-
U2 - Beta(0.7, 0.3)	5.2	6.6	9.6	6	2.3	-
U2 - Beta(1, 1)	5.2	6.6	9.9	5.9	2.1	-
U3 - Beta(0.3, 0.7)	5.6	7.2	12.4	3.9	0.5	-
U3 - Beta(0.5, 0.5)	5.5	6.8	9.9	5.7	1.9	-
U3 - Beta(0.7, 0.3)	5.6	6.8	9.4	5.9	2	-
U3 - Beta(1, 1)	5.8	6.9	9.6	5.7	1.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.6	4.3	1.6	0.2	-
U2 - Beta(0.5, 0.5)	0.9	3.7	3.1	2.1	0.7	-
U2 - Beta(0.7, 0.3)	0.9	3.7	3	2.1	0.7	-
U2 - Beta(1, 1)	1	3.6	3.1	2.1	0.7	-
U3 - Beta(0.3, 0.7)	1.2	3.7	4.2	1.4	0.2	-
U3 - Beta(0.5, 0.5)	1.1	3.8	3.1	2	0.6	-
U3 - Beta(0.7, 0.3)	1.1	3.8	2.9	2.1	0.7	-
U3 - Beta(1, 1)	1.2	3.8	3	2	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039.5
U2 - Beta(0.5, 0.5)						1053.7
U2 - Beta(0.7, 0.3)						1055.8
U2 - Beta(1, 1)						1053.8
U3 - Beta(0.3, 0.7)						1035.4
U3 - Beta(0.5, 0.5)						1052
U3 - Beta(0.7, 0.3)						1051.6
U3 - Beta(1, 1)						1048.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.6	70.5	19.1	1.6	0.1	0.1
U2 - Beta(0.5, 0.5)	7.7	42.1	34.9	12.7	2.5	0.1
U2 - Beta(0.7, 0.3)	10.2	42	30.8	14.3	2.6	0.1
U2 - Beta(1, 1)	8.3	41.8	35.7	11.9	2.2	0.1
U3 - Beta(0.3, 0.7)	11.5	72.8	14.2	1.4	0	0.1
U3 - Beta(0.5, 0.5)	11.1	47.7	31	8.4	1.7	0.1
U3 - Beta(0.7, 0.3)	12	47.5	27.7	11	1.7	0.1
U3 - Beta(1, 1)	12.6	49.7	28.6	7.7	1.3	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	17.4	3.6	0.5	0	-
U2 - Beta(0.5, 0.5)	4.4	12.1	8.6	3.8	1	-
U2 - Beta(0.7, 0.3)	4.4	9.3	8.7	5.5	2	-
U2 - Beta(1, 1)	4.4	12.2	8.6	3.7	1	-
U3 - Beta(0.3, 0.7)	8.8	17.7	3	0.3	0	-
U3 - Beta(0.5, 0.5)	4.7	13.1	8	3.3	0.8	-
U3 - Beta(0.7, 0.3)	4.6	10.9	8.5	4.7	1.2	-
U3 - Beta(1, 1)	4.6	13.4	7.9	3.3	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	6.4	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	0.5	4.8	2.4	1.2	0.3	-
U2 - Beta(0.7, 0.3)	0.5	3.9	2.3	1.6	0.6	-
U2 - Beta(1, 1)	0.5	4.8	2.4	1.1	0.3	-
U3 - Beta(0.3, 0.7)	2.6	6.6	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	0.6	5.2	2.2	1	0.2	-
U3 - Beta(0.7, 0.3)	0.6	4.4	2.3	1.4	0.4	-
U3 - Beta(1, 1)	0.6	5.3	2.2	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1094.3
U2 - Beta(0.5, 0.5)						1148.4
U2 - Beta(0.7, 0.3)						1165.5
U2 - Beta(1, 1)						1149.2
U3 - Beta(0.3, 0.7)						1091.1
U3 - Beta(0.5, 0.5)						1142.4
U3 - Beta(0.7, 0.3)						1154.1
U3 - Beta(1, 1)						1142.8
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.7	39.9	16.2	5.9	0.7	24.6
U2 - Beta(0.5, 0.5)	14.3	40.6	16.4	4.6	1	23.1
U2 - Beta(0.7, 0.3)	13.7	40	16.7	5.2	1.3	23.1
U2 - Beta(1, 1)	14.9	39.8	16.3	4.6	1.3	23.1
U3 - Beta(0.3, 0.7)	13.5	42.2	13.5	4.5	0.6	25.7
U3 - Beta(0.5, 0.5)	15.8	42.6	13.3	4.5	0.6	23.2
U3 - Beta(0.7, 0.3)	17	40.8	12.8	4.5	0.9	24
U3 - Beta(1, 1)	18	40.1	12.4	4.6	0.7	24.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	11.6	5	1.9	0.3	-
U2 - Beta(0.5, 0.5)	8.1	10.2	5.3	2.9	0.8	-
U2 - Beta(0.7, 0.3)	8.1	10	5.1	3.2	1	-
U2 - Beta(1, 1)	8.1	10.2	5.3	2.9	0.8	-
U3 - Beta(0.3, 0.7)	8.7	11.9	4.5	1.6	0.2	-
U3 - Beta(0.5, 0.5)	8.4	10.4	5.1	2.7	0.7	-
U3 - Beta(0.7, 0.3)	8.3	10.1	5.1	3	0.8	-
U3 - Beta(1, 1)	8.5	10.3	4.9	2.8	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5.3	1.8	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.8	4.9	1.7	1.1	0.3	-
U2 - Beta(0.7, 0.3)	1.8	4.9	1.6	1.2	0.3	-
U2 - Beta(1, 1)	1.8	5	1.7	1.1	0.3	-
U3 - Beta(0.3, 0.7)	2.2	5.5	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2	5	1.6	1	0.2	-
U3 - Beta(0.7, 0.3)	1.9	5	1.5	1.1	0.3	-
U3 - Beta(1, 1)	2	5	1.5	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						943.6
U2 - Beta(0.5, 0.5)						959.2
U2 - Beta(0.7, 0.3)						962.1
U2 - Beta(1, 1)						960.7
U3 - Beta(0.3, 0.7)						936.9
U3 - Beta(0.5, 0.5)						955.6
U3 - Beta(0.7, 0.3)						958.9
U3 - Beta(1, 1)						954.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.3 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12	49.8	14.2	2.7	0.3	21
U2 - Beta(0.5, 0.5)	13	50.2	13.5	2.8	0.4	20.1
U2 - Beta(0.7, 0.3)	14	49.4	13.8	3	0.5	19.3
U2 - Beta(1, 1)	14.9	47.6	14.7	2.7	0.6	19.5
U3 - Beta(0.3, 0.7)	15.5	50.9	12.5	1.7	0.2	19.2
U3 - Beta(0.5, 0.5)	16.1	50.9	11.3	2	0.2	19.5
U3 - Beta(0.7, 0.3)	17.3	49.2	11.7	2.4	0.2	19.2
U3 - Beta(1, 1)	19	46.7	12.1	2.5	0.2	19.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8	12.9	5.1	1.7	0.3	-
U2 - Beta(0.5, 0.5)	8	11.5	5.3	2.5	0.7	-
U2 - Beta(0.7, 0.3)	7.9	11.3	5.4	2.7	0.7	-
U2 - Beta(1, 1)	7.9	11.5	5.4	2.5	0.7	-
U3 - Beta(0.3, 0.7)	8.4	13.2	4.8	1.5	0.2	-
U3 - Beta(0.5, 0.5)	8.3	11.5	5.2	2.4	0.6	-
U3 - Beta(0.7, 0.3)	8.2	11.5	5.2	2.5	0.6	-
U3 - Beta(1, 1)	8.3	11.7	5	2.4	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5.8	1.8	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.8	5.5	1.7	1	0.2	-
U2 - Beta(0.7, 0.3)	1.8	5.4	1.6	1	0.3	-
U2 - Beta(1, 1)	1.8	5.5	1.7	1	0.2	-
U3 - Beta(0.3, 0.7)	2.1	6	1.7	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2	5.5	1.6	0.9	0.2	-
U3 - Beta(0.7, 0.3)	2	5.5	1.6	1	0.2	-
U3 - Beta(1, 1)	2	5.6	1.6	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						967.5
U2 - Beta(0.5, 0.5)						977.4
U2 - Beta(0.7, 0.3)						977.7
U2 - Beta(1, 1)						976.9
U3 - Beta(0.3, 0.7)						968.5
U3 - Beta(0.5, 0.5)						976.4
U3 - Beta(0.7, 0.3)						977.7
U3 - Beta(1, 1)						975.5